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BEMM466 Business Project

“Using Machine Learning Models as an Early Warning System to Predict Financial Distress for the UK Retail SMEs”

Prepared by Titova Vladislava

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Abstract

Although many studies have explored financial distress prediction, the topic still remains highly relevant because of constantly changing global trends and unique local conditions that motivate the development of context-specific models. This study aims to identify the most effective machine learning model for distinguishing between financially distressed and healthy companies within the UK retail small and medium enterprises. The analysis was based on secondary data, collected from the Orbis database, covering 1,000 UK retail SMEs between 2020 and 2024. Logistic Regression, Decision Tree, Random Forest, SVM, ANN, and LightGBM models were trained and tested with cross-validation and hyperparameter tuning. Results showed that Random Forest outperformed other models, achieving an accuracy of 0.840 for one year prior to financial distress model and 0.906 for two years prior to financial distress model, offering the best balance between recall and precision. This study contributes to financial distress research by providing a context-specific, explainable model that is tailored to UK retail SMEs, with practical explanations for key stakeholders.

Executive summary

Predicting financial distress has long been an area of interest for both academics and practitioners. However, the development of a universal model remains challenging, as business conditions are constantly changing and shaped by national and regional factors. This highlights the need for context-sensitive approaches that reflect the unique environment in which firms operate. In this regard, developing an accurate machine learning model is particularly relevant for the UK. Thus, recent reports indicate that the number of businesses in “critical” distress increased by more than 50 percent between the third and fourth quarters of 2024, with retail and other consumer-facing industries being among the most affected. Another distinctive feature of the UK business sector is that a substantial share of its businesses are SMEs, which make a significant contribution to the country’s economic output and employment. Therefore, developing a robust, interpretable, and context-specific early warning system is crucial to help to inform stakeholders of UK retail SME’s decision-making and ensure resilience within the sector. Consequently, the aim of this study is to identify the most effective machine learning model for predicting financial distress among UK retail SMEs. This aim was addressed through three main objectives - identifying the most relevant financial features for predicting distress, determining the optimal prediction horizon and evaluating the performance of different machine learning models in order to recommend the most suitable option.

The analysis was based on a dataset of UK retail SMEs, extracted from the Orbis database for the years 2020 to 2024, a period marked by the effects of the COVID-19 pandemic, which caused disruptions in supply chains and large changes in consumer behaviour. From an initial pool of more than 60,000 companies, a random sample of 1,000 was selected for the analysis. Financially distressed firms were defined using an accounting-based approach. Companies were classified as distressed if they experienced negative operating cash flow or a current ratio below 1. The independent variables used in the study were drawn from available financial ratios covering profitability, liquidity, solvency, market performance, and firm size. This includes profit margin, return on assets, profit on invested capital, profit per share, current ratio, cash

flow-to-assets ratio, debt-to-assets ratio and natural logarithm of total assets. Six machine learning models were selected for evaluation - Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Network (ANN), and LightGBM. Data preprocessing involved cleaning missing values, handling outliers, standardisation and addressing class imbalance through the Synthetic Minority Over-sampling Technique (SMOTE). Models were trained and validated with hyperparameter tuning and 5-folds cross-validation. Their performance was assessed using accuracy, precision, recall, and F-score metrics.

The results show that no single model outperformed all others across every metric. Logistic Regression achieved the highest recall, identifying up to 92.3 percent of distressed firms when predicting two years in advance. However, this closely connected with a cost of very low precision, meaning that many healthy firms were incorrectly classified as distressed. While this outcome minimises the risk of overlooking true distress cases, it creates inefficiencies and resource burdens, particularly for SMEs that can have limited budgets. At the same time, Random Forest demonstrated the best balance across all metrics, achieving the highest accuracy (0.906 two years before distress) and the strongest F-score, reflecting a more reliable trade-off between precision and recall. LightGBM also performed strongly, though slightly below Random Forest. An interesting finding was that ensemble tree-based models, such as Random Forest, LightGBM, and Decision Tree, tended to perform better when predicting distress two years in advance compared to one year. This suggests that these models are prone to capture medium-term patterns that may be less visible in the immediate short term, and can provide valuable early signals for timely intervention.

Model interpretability was addressed by using feature importance rankings and the application of explainable AI techniques such as SHAP (SHapley Additive exPlanations). The analysis of feature importance revealed that the current ratio and debt-to-assets ratio were consistently the most influential predictors of financial distress across both horizons. Global explanations confirmed that current ratio is the most significant feature in predicting financial distress as well as provided the direction of the feature influence. Overall, the results are quite predictable and do not contradict the common influence of

financial ratios. However, both models indicate that high level of debt-to-asset ratio decreases the probability of financial distress. This may be because a moderately high ratio indicates that a company has access to credit markets, providing it with greater financial flexibility, which helps to avoid financial distress. The local explanations can be quite valuable tool as well, since they can provide the information, why a model made a specific prediction for the specific company. This increases the transparency and would help companies to tailor the corrective measures. It is also important to mention that the explanations gave an interesting results. For example, depending on the certain level of deviation and in intereaction with other variables, the current ratio had the different effect on the model prediction.

The findings of this study can carry important implications for managers, investors, and creditors operating in the UK retail SME. For managers, the Random Forest model supported by SHAP explanations provides a practical decision-support tool, enabling proactive monitoring of financial health and better allocation of resources. Importantly, the results highlight that liquidity and solvency measures, particularly the current ratio and debt-to-assets ratio, are consistent indicators of financial vulnerability. Given that retail SMEs typically operate with thin margins, high inventory levels, and frequent reliance on short-term financing, early deterioration in these ratios can quickly escalate into severe financial distress. The force plot analysis further demonstrates that interactions between features matter. It highlights that managers should not rely on ratios in isolation, but instead consider the broader financial profile of their firms when making strategic decisions. Investors can benefit using the result of this study as accurate predictions of financially distressed firms can reduce risks and transaction costs by avoiding companies that may appear profitable but are structurally unstable. For creditors, the model offers a more nuanced basis for credit risk assessment, allowing them to differentiate between firms that are temporarily under pressure but resilient, and those genuinely at risk of default. By adopting these insights, stakeholders can move beyond surface-level financial indicators and instead evaluate SMEs through a more comprehensive and context-specific lens, ultimately supporting more sustainable growth and financial stability within the sector.

It is worth to mention that the study is not without limitations. The dataset covers only UK retail SMEs between 2020 and 2024, limiting the generalisability of results to other sectors or time periods. Future economic shocks could significantly affect predictive patterns, meaning that models must be regularly estimated with updated data. The analysis was also restricted to accounting-based variables, which, while informative, may not capture other aspects of firm performance. Incorporating additional indicators, such as macroeconomic, corporate governance, or textual data, could improve predictive abilities of models. In addition, combining accounting-based indicators with legal records to identify financially distressed companies could help build a classification framework that better reflects real-world conditions. Finally, this study focuses on traditional models, however, future research could explore the development of hybrid approaches to further enhance predictive power.

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1. Introduction

The topic of predicting company's financial distress has long attracted the interest of academics and practitioners, including key company stakeholders such as managers, investors, and creditors. Despite the constantly growing body of literature focused on developing models to predict financial distress, the topic continues to be highly relevant due to the rising demand for highly accurate and interpretable models. Nevertheless, developing a universal model is challenging, as business conditions are constantly evolving and shaped by national and regional factors which can drive ongoing research into more context-sensitive approaches that can reflect the unique environments in which companies operate.

It is worth noting that the development of the accurate machine learning model would be highly relevant in the context of the UK. According to recent reports from insolvency experts, there has been a sharp increase in UK businesses identified as being in "critical" financial distress (Jack & Edwards, 2025). For example, the number of such cases rose by 50.2% between the third and fourth quarters of 2024 - from 31,201 to 46,853 instances. The sectors most affected were consumer-facing industries, such as food & drug retail, hotels & accommodation, leisure and cultural activities (Beggies Traynor Group, n.d.). While firm closures are a natural part of economic cycles, atypical exit of companies driven by financial distress can have severe negative effects on the broader economy (Sehgal et al., 2021). For instance, the retail sector alone generated £111.6 billion in output in 2024, representing 4.5% of the UK's total output (Ward, 2025). Thus, improving early detection of financial distress in this sector could help mitigate significant economic losses.

Additionally, modeling financial distress for SMEs can be particularly significant, as these businesses are less frequently audited, making them more vulnerable to undetected risks. Moreover, SMEs play a vital role in the UK economy, representing 99.9% of the business population at the start of 2024 and acting as a foundation for local communities through job creation and economic contribution (Parker, 2023; Ward, 2025). Having access to an accurate prediction model can help SME owners take preventive action. Therefore, a

reliable model for predicting SME financial distress would deliver widespread economic and social benefits.

The aim of this research is to identify a machine learning model that can accurately distinguish between financially distressed and healthy companies in the UK SME retail companies. To support this aim, the following objectives and questions have been identified that are presented in Table 1.

Table 1. Identified research objectives and corresponding questions.

Research objective	Research questions
1. Identify the most relevant features of financial distress	1. Which variables have been most commonly used in prior research to predict corporate financial distress? 2. Are there notable differences in the average values of identified indicators between distressed and non-distressed companies?
2. Determine the optimal prediction time horizon	1. Which time horizon provides better model performance based on evaluation metrics?
3. Identify the best machine learning model	1. Which machine learning models are most suited to achieve the aim of current study? 2. Which machine learning models have shown the strongest performance based on evaluation metrics?

The outcomes of this project will be valuable to a range of stakeholders, including managers, investors, and creditors (Mselmi et al., 2017; Sreedharan et al., 2020). For managers, an early warning system provides a practical tool to monitor the financial health of their businesses and take timely corrective actions, such as improving liquidity management, restructuring debt or adjusting operational strategies, before distress escalates into insolvency. For investors, the ability to identify early signs of financial weakness helps to make more informed decisions about allocating capital, reducing the risk of investing in firms that may fail, while also highlighting more resilient SMEs with strong financial indicators. Creditors, including banks and other financial institutions, can benefit by incorporating such predictive models into their credit risk assessments,

allowing them to better differentiate between healthy and distressed borrowers. This reduces the likelihood of loan defaults and enhances their ability to design lending terms that reflect the true risk profile of SMEs. By providing tailored, data-driven insights, the outcomes of this research can therefore contribute to more sustainable decision-making and improved financial stability across the UK retail SME landscape.

In summary, this research seeks to contribute to the ongoing academic and practical discussion on financial distress prediction. While some studies have examined financial distress in UK companies, few have focused specifically on the UK's SME retail sector. Furthermore, much of the existing research predates major economic disruptions such as Brexit, the COVID-19 pandemic and the current cost-of-living crisis, all of which have significantly altered the business landscape. This highlights the need for updated models trained on recent data that reflect today's economic conditions.

The rest of the following paper is organised as follows. Section 2 presents the literature review, which helps to identify the gap in existing research and supports the initial selection of variables most commonly used in prior studies to predict corporate financial distress. This contributes to answering question 1 of the first objective. Section 3 provides an extensive overview of the methodology, including data sources, data collection, description of independent and dependent variables, justification and description of the chosen models, as well as the data workflow and limitations of the current study. This section helps to address question 2 of the first objective. Section 4 answers the second objective and question 2 of objective 3 by identifying the best-performing model and providing an explanation for it to make it more transparent to the key stakeholders and applicable to the real - world tasks.

2. Literature review

2. 1. Identifying target variable

The initial step in modelling financial distress is to clearly define the concept in order to determine which companies can be classified as “distressed”. A key challenge in this process is the broad nature of the term, which has led to the absence of a universally accepted definition. Broadly, financial distress refers to a company's inability to generate

sufficient operating cash flow to meet its financial obligations, often requiring corrective action (Gruszczyński, 2020).

It is worth mentioning that different studies adopt various criteria to classify financial distress. Some researchers use bankruptcy status as a proxy, while others rely on legal or institutional definitions (Aydin et al., 2022; Tinoco & Wilson, 2013). For instance, Mselmi et al. (2017) define distress based on judicial declarations, whereas Zizi et al. (2021) consider firms as distressed if they fail to meet credit obligations for over 90 days. However, in order to facilitate early detection, many studies employ financial indicators to identify distress before formal insolvency occurs (Abrahamsen et al., 2024; Martono & Ohwada, 2023; Sreedharan et al., 2020; Tinoco & Wilson, 2013; Sanjay & Mishra, 2020). For instance, Abrahamsen et al. (2024) classify firms as distressed when the interest coverage ratio is below 1.5 and the current ratio is under 1, while Martono and Ohwada (2023) use negative operating cash flow. Tinoco and Wilson (2013) propose a hybrid model combining three criteria: (1) EBITDA below financial expenses for two consecutive years, (2) a two-year decline in market value, and (3) formal default—merging legal, market-based, and accounting-based definitions to improve classification accuracy.

2.2. Independent variables

Identifying key predictors of distress is the next important step of model construction. Beaver's early research showed financial ratios have strong explanatory power. Building on that, Altman developed multivariate models confirming the predictive value of financial ratios (Mselmi et al., 2017; Sehgal et al., 2020). Consequently, financial ratios remain widely used in recent research. However, as businesses and data environments become more complex, many scholars note that financial ratios, that are based on historical, quantitative data, may have limited predictive power as they may contain outdated information (Song et al., 2022; Colff & Brummer, 2019; Tinoco & Wilson, 2013). Recent studies advocate combining financial variables with non-financial indicators, such as management quality, social responsibility factors, and macroeconomic variables (Song et al., 2022; Colff & Brummer, 2019; Judijanto et al., 2024). For example, Song et al. (2022) include management structure indicators such as the number of directors, the number of

independent directors, the number of shares held by executives, and whether the roles of chairman and CEO are separated.

2.3. Models

When it comes to the modelling step, financial distress prediction methods broadly fall into statistical and machine learning (ML) models. Statistical models are considered as easy to use and interpret, but rely on assumptions such as linearity, normality, and variable independence, which may not hold in real-world data (Song et al., 2023; Colff & Brummer, 2019). Due to increasing data complexity, many recent studies apply machine learning methods, which can offer higher accuracy, flexibility, and the ability to process complex, multidimensional data (Zizi, 2021; Sreedharan et al., 2020).

ML models can be broadly categorized into single and ensemble classifiers. Single classifiers such as Decision Trees (DT), Support Vector Machines (SVM), and Artificial Neural Networks (ANN) are widely used for their speed and performance. At the same time, ensemble classifiers including voting-based, bagging-based, and boosting-based models are gaining more attention for their superior predictive capabilities (Song et al., 2023). For instance, Martono and Ohwada (2023) compare the performance of 4 models – Logistic Regression (LR), Decision Tree (DT), LightGBM and KNN. Empirical results demonstrates the accuracy range between 0.60 and 0.87, where the highest accuracy value of 0.87 was achieved by LightGBM algorithm for the scenario of 1 year prior to the financial distress. This aligns with the result of Zhang et al. (2022), where LightGBM model shows the best performance across all the presented metrics. However, ensemble models do not always perform best. Sreedharan et al. (2020) highlights that despite the fact that Majority Voting classifier showed the highest average accuracy and that it outperforms the results of AdaBoost and RF, it could not outperform the average accuracy of Neural Network classifier. Similarly, comparison between statistical and machine learning also show contradictory outcomes. Thus, Zizi et al. (2021) found that Logistic Regression outperformed ANN, with accuracy of 95% versus 88.33%. Therefore, no classifier consistently performs best and the result might be highly dependent on dataset characteristics.

2.4. Discussion on explainable AI

As mentioned earlier, machine learning models are increasingly popular for financial distress prediction due to their ability to handle complex, non-linear relationships and their flexibility compared to traditional statistical models, which often rely on restrictive and unrealistic assumptions. However, some ML models function as “black boxes”, offering limited transparency and interpretability. This lack of clarity can lead to a reduced trust from the key stakeholders and hinder decision-making. As a result, there is growing concern about the importance of providing model interpretation to facilitate stakeholders’ understanding and confidence (Judijanto et al., 2024; Zhang et al., 2022). To address this issue, explainable AI (XAI) tools have been developed to improve model transparency. Common methods include SHapley Additive exPlanations (SHAP), Partial Dependence Plots (PDP), and counterfactual explanations, which help to provide global and local explanations and give some practical insights to the stakeholders (Zhang et al., 2022; Liu et al., 2022).

In summary, the topic of financial distress is certainly comprehensive and includes different aspects to analyse, such as identifying distressed companies, selecting significant features and models and providing the interpretation of result. Moreover, the model performance is highly context-dependent, which motivates to farther exploration of this topic and finding the best-performing model on specific data. Thus, For UK retail SMEs that face unique risk profiles and economic pressures customized, interpretable, and robust ML models are essential.

3. Methodology

3.1 Data ethics

This study will ensure ethical data processing by following the principle of data minimisation, meaning that only information essential to the research purposes will be collected. Thus, only quantitative financial data that aligns with the original research objectives will be downloaded.

In line with good research practice, data security and storage will be prioritised throughout the project. All data will be stored securely on OneDrive cloud environment, with access restricted to individuals directly involved in the project.

Transparency in data handling will also be ensured. Each step of the data processing will be clearly documented, including Python code used for data importing, cleaning, model testing and training, and showing final results. This will help to guarantee that the work with the data remains fair, traceable and ethical.

The principle of data retention and disposal will also be observed. The dataset will be retained only for as long as necessary to complete the dissertation and assessment process. After the completion of the degree, all data will be permanently deleted to prevent any further use or disclosure.

3.2 Data source

To address the research aims and objectives, this study will utilise secondary data obtained from the Orbis database, provided by Bureau van Dijk. Orbis is a reputable source that is widely recognised as one of the most comprehensible and reliable sources of company-level data that integrates information on over 580 million companies globally. The database consolidates information from official sources such as company registries, credit agencies and regulators.

By relying on a database that follows rigorous validation procedures and integrates multiple official data sources, this research ensures that the analysis is based on robust and high-quality evidence. This can help to strengthen both the credibility of the findings and their practical applicability to real-world business context.

3.3 SME definition

The current research specifically targets small and medium-sized enterprises (SMEs), operating within the UK retail sector. In order to ensure a consistent and transparent definition, the study adopts the official classification of SMEs provided by the UK

government. Thus, according the Government Commercial Function (2023), a firm can be qualified as an SME if it satisfies the following criteria:

- It employs fewer than 250 people;
- An annual turnover of £44 million or less or total balance sheet of £38 or less;

These parameters along with the NACE 47 (Retail trade, except of motor vehicles and motorcycles, which covers the full range of retail activities conducted in stores, via mail or online) industry classification will form the basis for the dataset collection.

3.4 Time horizon

In terms of the time horizon, it was decided to focus the analysis on the period between 2020 and 2024. The specified timeframe was selected, since it reflects the most relevant trends in the UK retail sector following the COVID-19 pandemic. It is important to mention that the pandemic acted as a major turning point that caused major disruptions of traditional business models, reshaped established supply chains, and accelerated structural changes, such as impetuous expansion of e-commerce, digitalisation and significant shifts in consumer preferences. As a result, the retail environment after 2020 is fundamentally different from the period before the pandemic, making earlier data less representative of the current realities faced by SMEs.

The inclusion of 2024 as the end point is motivated by the intention to capture the most recent and complete set of available financial data. Using 2024 helps to ensure that the dataset reflects the latest market conditions, including post-pandemic recovery, inflationary pressures and the overall economic climate that influenced retail trade in recent years.

Overall, focusing on 2020-2024 period allows this study to analyse both pandemic-driven transformations and the subsequent adjustment in the sector, providing a balanced and up-to-date foundation for analysis.

3.5 Independent variables

Based on the analysis of literature devoted to the topic of predicting financial distress, it can be concluded that accounting-based variables are still widely recognised in the literature as reliable predictors of financial distress. Therefore, this study is also going to adopt an accounting-based approach.

Although the Orbis database provides relatively limited set of financial variables, nonetheless, available features represent the core aspects of profitability, liquidity, solvency, size and overall financial performance, making them relevant for the objectives of this study (see the Table 1). A closer examination of the selected ratios is presented below.

Table 2. Features, adopted in the study

Themes	Ratio	Formula
Profitability	Profit Margin	$\frac{\text{Net Profit}}{\text{Revenue}}$
Profitability	Return on Asset	$\frac{\text{Net Profit}}{\text{Total Assets}}$
Profitability	Profit on invested capital	$\frac{\text{Profit} - \text{Preferred Dividends}}{\text{Weighted average common shares}}$
Market related	Profit per share	$\frac{\text{Net profit}}{\text{Total invested capital}}$
Liquidity	Current ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$
Solvency	Debt – to – Assets	$\frac{\text{Total Liabilities}}{\text{Total Assets}}$
Liquidity	Cash flow – to – Assets	$\frac{\text{Cash flow}}{\text{Total Assets}}$
Size	Log(Total Assets)	

Profitability aspect of the company is presented by 3 ratios. Firstly, Profit Margin measures the percentage of revenue that translates into profit, reflecting operational

efficiency (The Perfect Answers, 2024; Segal, 2025). A consistently low or declining margin suggests limited financial resilience and may potentially increase the likelihood of financial instability and financial distress. At the same time, ROA captures how effectively a company uses its assets to generate profit. Thus, a persistently low ROA signals inefficient resource use and is considered as a common symptom of distress (Skilling Writers Team, 2024). Profit on Invested Capital assesses how efficiently capital is employed to generate earnings, with declines indicating weakening efficiency or poor investment decisions.

Profit per Share, which is classified as a market related ratio, links firm performance to shareholder value, as it reflects the portion of a company's profit attributable to each outstanding share. Declining levels can signal weakening profitability and growth prospects, erode investor confidence, restrict access to external financing, and ultimately increase the likelihood of financial distress. This ratio is included in the model as it incorporates external market-related information, capturing factors beyond those reflected in purely accounting-based financial ratios.

Another widely applied measure is the Current Ratio, defined as current assets divided by current liabilities. Usually, a ratio below 1 indicates liquidity concerns, as liabilities exceed assets, raising distress risk (Fernando, 2025). Given its role in assessing short-term solvency, it is included in the model. Another liquidity measure is Cash Flow-to-Assets Ratio, which captures how effectively a firm generates cash flow from its asset base. As cash flow is less influenced by accounting policies than profit measures, this ratio can be a robust indicator of a company's ability to sustain operations. Higher values imply stronger financial health, while low or negative values indicate elevated distress risk (FasterCapital, 2025).

As to see the solvency indicator influence, Debt-to-Asset Ratio is included in the model, which evaluates the share of assets financed through debt. Higher values of this ratio may imply elevated leverage and financial vulnerability, while lower values reflect stability and reduced insolvency risk. Excessive debt often precedes financial distress, making this ratio a critical predictor.

Finally, following prior research, natural logarithm of Total Assets is included as a proxy for firm size. Larger firms generally benefit from scale, resources, and access to financing, reducing distress vulnerability. Smaller firms, by contrast, face greater risks due to limited assets, weaker financing options, and lower resilience.

3.6 Dependent variable

When defining the dependent variable, this study adopts an accounting-based approach to identify financially distressed companies. Rather than relying on the legal status of a company being bankrupt, the focus is placed on financial indicators that serve as early warning signals of distress. The motivation for this decision is that bankruptcy represents the final stage of financial failure, while accounting-based measures allow researchers and practitioners to detect vulnerability at earlier stages, when corrective actions may be implemented to prevent financial distress.

Accordingly, the indicators chosen for this study are directly linked to a firm's solvency and liquidity position, reflecting its ability to meet obligations as they arise. A company will be classified as financially distressed if it exhibits negative operating cash flow or a current ratio below 1. Thus, negative cash flow indicates that the organization is not generating sufficient internal funds to support its operations, creating immediate liquidity pressure. Moreover, a current ratio below 1 signals that current liabilities exceed current assets, suggesting that the firm lacks adequate level of resources to cover its debts.

3.7 Chosen ML models

In this study it is intended to compare the following models – Logistic Regression, ANN, SVM, Decision Tree, Random Forest and LightGBM. Logistic Regression was chosen as a benchmark model due to its simplicity and interpretability. However, LR requires the linear relationship between target and independent variables, is sensitive to outliers, which may limit its performance (Mselmi et al., 2017; GeeksforGeeks, 2025). Therefore, ML models are also included.

As to single classifiers, it is intended to use Artificial Neural Network, Support Vector Machine and Decision Trees. DT was selected for its low complexity and high

interpretability. ANN and SVM, though considered “black box” models, they don’t rely on strict assumptions and are able to capture linear and non-linear patterns in data. The disadvantage of these models is that they offer low interpretability as well as there is difficulty fixing the parameters of the model, which requires hyperparameter tuning.

As to ensemble classifiers, RF and LightGBM were chosen for the analysis and represent bagging and boosting approaches, respectively. RF is widely used due to its ability to output feature importance and its robustness to overfitting (Smith, 2024). LightGBM was selected over GBDT and XGBoost as it addresses their limitations - it is faster, uses less memory, and is less computationally intensive while maintaining high performance (Qian et al., 2022).

3.7.1 Logistic Regression model

Logistic regression is a widely used statistical and machine learning model for predicting binary outcomes, such as distinguishing between financially distressed and non-distressed firms. In contrast to linear regression, which predicts continuous values, logistic regression estimates the probability that an observation falls into a particular category by applying the logistic (sigmoid) function. This ensures that predicted values lie between 0 and 1. The model assigns weights to independent variables, and the sign and magnitude of these coefficients indicate the direction and strength of their influence on the likelihood of the event occurring.

3.7.2 Decision Tree

Decision Tree (DT), a supervised machine learning algorithm that can be applied to both classification and regression tasks. Decision trees operate by recursively splitting the dataset into smaller and more homogeneous subsets based on the most significant features, thereby forming a hierarchical tree-like structure. The logic behind this process is to partition the data according to rules that maximize the separation between classes, making the model intuitive and highly interpretable.

The structure of the tree consists of three main components - nodes, branches, and leaves. The root node is the entire dataset before any split is made. From there, the

algorithm identifies the most informative feature, based on criteria such as Gini impurity, and divides the data accordingly. These splits are represented as internal nodes, which correspond to decision points where the data is divided into smaller groups. The branches are the pathways that connect nodes, representing the outcomes of each decision rule, for example, whether a firm's financial ratio exceeds a certain threshold. Finally, the leaf nodes the final classification outcomes, such as whether a firm is predicted to be financially distressed or healthy (Koli, 2023).

3.7.3 ANN

Another important model applied in classification problems is the Artificial Neural Network (ANN). ANNs are a class of machine learning algorithms inspired by the structure and functioning of the human brain. They are designed to recognize complex patterns by learning non-linear relationships between inputs and outputs. Unlike logistic regression or decision trees, which rely on relatively simple mathematical or rule-based structures, ANNs are capable of approximating highly complex patterns, making them particularly powerful for tasks where the relationship between variables is not strictly linear.

An ANN is typically organized into layers of interconnected nodes, also known as artificial neurons. The input layer consists of nodes that represent the features of the dataset. In the context of financial distress prediction, these input nodes would correspond to the financial indicators of firms, such as leverage ratios, liquidity ratios, profitability measures, or market-based variables. These inputs are then passed through one or more hidden layers, where the network processes and transforms the information using weighted connections and activation functions. Each hidden layer is designed to capture increasingly abstract relationships within the data. Finally, the output layer produces the final prediction. In this study, the output layer would contain a single node that represents the probability of a firm being financially distressed.

3.7.4 SVM

Support Vector Machine or SVM is a machine learning algorithm that is also well-suited for classification tasks, especially when the relationship between variables is complex and

not easily captured by linear models. The main idea behind SVM is to find the optimal hyperplane that best separates the data points of different classes in the feature space. This hyperplane is chosen such that it maximises the distance between the hyperplane and the nearest data points from each class, known as support vectors.

In the context of financial distress prediction, the input layer of an SVM corresponds to the financial variables of firms, such as leverage, liquidity, profitability, and market indicators. These variables are transformed into points in a multidimensional feature space. The algorithm then searches for the hyperplane that separates distressed firms from healthy ones with the greatest possible margin. The output is a classification decision: whether a firm is predicted to be financially distressed or not.

3.7.5 Random Forest

Another widely applied model for classification tasks is the Random Forest (RF) algorithm. Random Forest is an ensemble learning method that combines the results of multiple trees to improve both accuracy and robustness. The central idea of Random Forest is to construct multiple decision trees on different bootstrapped subsets of the data and then aggregate their predictions, typically through majority voting in classification problems. This process reduces the risk of overfitting associated with individual decision trees, while at the same time enhancing predictive power and generalization.

3.7.6 LightGBM

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework that builds decision trees sequentially to minimize prediction errors. One of the main features of the LightGBM is that it grows trees leaf-wise rather than level-wise, focusing on nodes with the greatest loss reduction. This strategy allows it to achieve higher accuracy with fewer iterations while handling large datasets efficiently.

3.8 Evaluation metrics

In order to compare different models and determine the most optimal one, this study will employ evaluation metrics such as accuracy, precision, recall, and F-score. These

metrics are among the most commonly used in the assessment of classification models, since each of them provides a unique perspective on the predictive performance of the model. It is important to mention that accuracy, recall and precision are calculated on the basis of the confusion matrix, which provides a detailed summary of how the predictions of a classification model align with the true classes. The confusion matrix is structured in a way that maps predicted outcomes against actual outcomes. Thus, in the context of the financial distress prediction, True Positives (TP) represent the number of distressed companies that are correctly identified by the model, whereas True Negatives (TN) denote the number of healthy companies correctly classified as non-distressed. False Positives (FP) occur when a healthy company is incorrectly flagged as distressed, which is known as a Type I error. Conversely, False Negatives (FN) capture the instances in which a distressed company is misclassified as healthy, which is referred to as a Type II error.

Each of the evaluation metrics is derived from these categories. Accuracy is defined as the ratio of correctly classified companies, both distressed and healthy, over the total number of observations (see equation 1). Despite the fact that accuracy is often considered as the most straightforward and intuitive measure, relying solely on accuracy can be misleading, especially on real-world data. Precision, on the other hand, measures the proportion of companies that the model predicts as distressed which are truly distressed (see equation 2). This metric reflects the model's ability to avoid false alarms, as a higher precision value indicates fewer Type I errors. Recall, sometimes referred to as sensitivity, measures the proportion of truly distressed companies that the model successfully identifies (see equation 3). In other words, recall captures the model's effectiveness in minimizing Type II errors, which is of critical importance in financial distress prediction. Finally, the F-score (see equation 4) represents the harmonic mean of precision and recall, providing a balanced measure that incorporates both metrics into a single value. This score is particularly useful when both types of errors carry significant consequences and when the objective is to balance the trade-off between capturing distressed companies and avoiding unnecessary false alarms.

It is important to emphasize that there is always a trade-off between reducing false positives and false negatives, and the prioritization of one type of error over the other depends heavily on the context of application. In the case of predicting financial distress, Type II errors are generally considered more harmful than Type I errors. Thus, if a model fails to detect a distressed company, management may lose the opportunity to undertake timely corrective actions. In contrast, Type I errors, while still problematic, result in healthy companies being mistakenly flagged as distressed, which may lead to unnecessary concerns, which can increase company expenses.

For retail small and medium-sized retail enterprises, the consequences of missing a financially distressed company are particularly severe, as SME's often have limited access to capital markets and fewer resources to withstand prolonged financial difficulties. Therefore, recall can be considered as a key metric, as it ensures that the majority of distressed firms are captured by the model. At the same time, a model that maximizes recall without consideration for precision may generate excessive false alarms, which could reduce the credibility of the system and lead to inefficient allocation of managerial attention and resources. For this reason, the F-score can be recommended as the most appropriate metric for evaluating models in this context, since it balances the need for high recall with the importance of maintaining reasonable precision. By optimizing for the F-score, management of retail SMEs can benefit from a more reliable and balanced tool for identifying financial distress at an early stage, enabling timely interventions and mitigating risks for both internal decision-makers and external stakeholders.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F - score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (4)$$

3.9 Data workflow

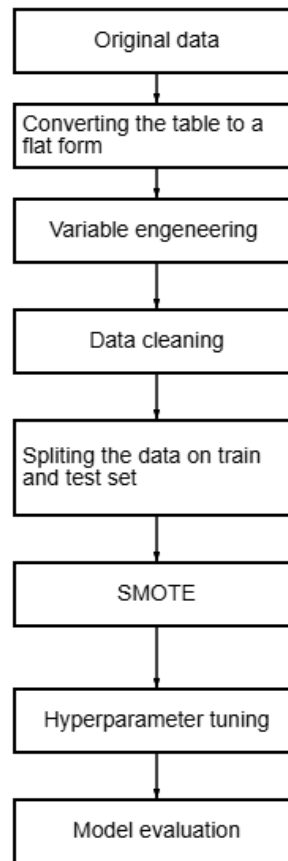


Fig.1. Data workflow

Figure 1 represents the universal data workflow to evaluate the chosen models that are adopted in the current study. As it can be seen from the figure, the first step is the data collection. Although the initial filtering of UK retail SMEs yielded over 60,000 companies, a random sample of 1,000 firms was selected for the analysis. This decision was made due to the limitations of the scope of the present research and to account for existing computational constraints associated with processing the complete dataset.

The data was downloaded in the form of the Excel file and was imported to the coding environment using *pandas* library. Since the initial data was presented as a wide table, it was essential to transform it into a flat table for the purpose of the current analysis. Even though the majority of financial ratios were already given, it was decided to calculate more ratios based on the data based on given absolute values. Moreover, two columns were

created that represented the one year before financial distress and two years before financial distress cases. It was done in order to compare models predicting distress one year before financial distress and two years prior to financial distress, because it can show the trade-off between higher accuracy and earlier warning. Thus, a one-year prediction might be more precise, but a two-year prediction gives more time to take action before problems become critical

After the feature engineering step, the initial cleaning data process was conducted, where all the missed values were deleted from the analysis. It is worth to mention that, overall there are two main approaches to handling missing data values which include imputation and deletion. Taking into account the fact that this research is focuses on the small and medium companies, it is expected that the amount of sufficient data is quite sufficient. Therefore, despite the fact that dropping rows with missing values have considerable drawbacks such as it reduces sample size, the imputed variables are not always reliable and also can reduce the variance of that data, that can lead to the biased results (Kartika, 2020; GeeksforGeeks, 2025). Consequently, it was decided to completely eliminate the rows with the missed values.

The following data cleaning process included checking the data for outliers. It is important to mention that checking for outliers is important, especially for models such as Logistic Regression, SVM and ANN. Tree-based models (DT, RF, LightGBM) are considered as more robust to outliers, but it was decided to use the same data without outliers for them for the uniformity of the process. As can be seen from figure 8 before adjusting the data for outliers (see the Appendix 1) the data was quite skewed, which could be seen from the sufficient amount of data points outside of the range of interquartile interval, which could possibly lead to the inaccurate result for the outlier-sensitive models. However, after dropping the missing values distributions of continuous variables became less skewed and became more normal (see figure 9).

The next step is filtering the features. After identifying the initial feature selection based on the literature review analysis, it was important to answer the question – are there notable differences in the average values of identified indicators between distressed and non-distressed companies. Answering this question helps to filter which variables can

have high predictive ability. It could be noted that the chosen database does not include big number of variables, it is still important to drop features that does not have high predictive power. This would help to acquire more accurate result, and could decrease the complexity of the model, which can lead to faster model evaluation process. Furthermore, it was decided to conduct a paired t-test when variances between groups are equal, and Welch's t-test when variances are unequal. These tests were conducted with the help of *stats* package from *scipy* library. It is also important to highlight that in order to estimate the homogeneity of the variances, the Levene's test, for the equal variance test, was conducted (Htoon, 2020).

After feature selection, the standartisation technique was applied for the remaining variables for the Ligistic Regression models and other ML models separately. It was done using the *StandardScaler* function that is offered by the *sklearn.preprocessing* library. This step is highly important for such models as LR, ANN and SVM, since these models rely on calculations based on distances and gradients, making them highly sensitive to the features with different scale. At the same time, it does not have any effect on the tree-based models, since their splits are based on the rank and ordering.

Figure 10 provides information on which test was conducted for each variable, along with the corresponding results that are based on the p-value score. Thus, when p-value is above the certain significance level, it means that it is not possible to reject the null hypothesis that the means of two groups are equal. As it could be seen from the table, only for Total_assets_In there is no significant difference (p-value is 0.33), which means that this variable is not informative and can be eliminated from the analysis.

When talking about feature elimination, it is important to mention that for Logistic Regression the further analysis of the chosen variables was conducted in order to eliminate the problem of potential multicollinearity, which lead to increase of the standard error of the variable coefficient estimate and can lead to the biased results (Siddiqui, 2019). Therefore, it was decided to look at the correlation between variables, in order to eliminate the highly correlated variables. Thus, figures 11 and 12 shows the correlation between the variables before and after the variable elimination. As it can be seen in figure 11, there is high correlation between significant amount variables. After the feature

elimination, there were only 3 variables left – current ratio, profit margin and profit per share, which represent different aspects of financial condition of the company. For other models, all the variables, apart from the `Total_assets_In`, were included in the evaluation, since they are less sensitive to the correlation between variables.

As shown in figure 1, the next step is dividing the dataset into train and test parts. The train part is needed to make models “learn” from the data, while the test part is devoted to evaluation of the model performance. This step was conducted by using the *train_test_split* module from the *sklearn.model_selection* library. The rule for the split was 80% of the data went for the training, while other 20% - for testing, which is generally common ratio for splitting datasets.

After the splitting the data, the SMOTE technique, that stands for the Synthetic Minority Over-sampling Technique, was utilised in order to address class imbalance for all the presented models. Figure 13 highlights that the data is highly imbalanced. Looking at the percentage of the distressed and non-distressed companies over the presented time horizon, it is seen that the number of companies suffering from the financial distress is significantly lower than the number of healthy companies, that can negatively affect the model performance as it can overlearn on the majority class data. SMOTE addresses this problem by generating synthetic samples for the minority class (GeeksforGeeks, 2024). It is also important to highlight that SMOTE is done only after splitting the data on the train and test set in order to avoid the problem of data leakage. When the synthetic samples are created based on the entire dataset, the train data would also contain the unreal data and can no longer be considered as real-world data. Eventually, that can lead to misleading model performance. This technique was done using *SMOTE* module from the *imblearn.over_sampling* library.

Finally, there was the step of model evaluation with hyperparameter tuning and cross-validation. Hyperparameter tuning was done in order to find the most optimal model parameters and reach the highest possible accuracy of the models. This was conducted by using the *GridSearchCV* package. Additionally, cross-validation technique was done to avoid overfitting, when model performs well only on the specific test set, that can affect the generalising abilities of the model. It was decided to use 5-folds cross validation, since

the given dataset is comparably large, while computational resources of the current study are limited. After finding the optimal parameters, models were trained on the trained data and tested on the test data. The results are presented in the Result section.

3.10 Limitations

Before applying the findings of this research in practice, it is important to acknowledge several limitations. First, the model was trained on data from UK retail SMEs covering the period 2020 to 2024, which may restrict its generalisability, particularly in the event of major future economic shocks. Second, the study relies exclusively on accounting-based variables. While these provide valuable insights, predictive performance could be enhanced in future research by incorporating additional features such as textual data, macroeconomic indicators, or governance-related variables. Third, the identification of financially distressed companies in this study was based on negative cash flows and a low current ratio. It could risk introducing a degree of circularity when these same indicators in one form or another are included as independent variables. Furthermore, not all firms experiencing negative cash flows or weak liquidity can be classified as financially distressed and vice versa. This approach was adopted because information on firms' legal status was not available, which may have led to an overstatement of the number of distressed companies. Future research could address this limitation by combining accounting-based measures with legal data, providing a classification framework more closely aligned with real-world conditions. Finally, this study adopted traditional models, however, the use of hybrid models that combine the strengths of different algorithms may further improve prediction accuracy and mitigate the weaknesses of individual classifiers.

4. Results

4.1 Models' performance evaluation

To address the research question of which machine learning model is the most effective for predicting financial distress, six algorithms were evaluated - Logistic Regression, Artificial Neural Network, Support Vector Machine, Decision Tree, Random Forest and LightGBM. In total, twelve models were assessed, with each algorithm applied to two

prediction horizons - one year and two years prior to distress. All models were implemented in Python and trained using cross-validation and hyperparameter tuning, employing libraries such as *LightGBM* and *scikit-learn* (for other models) to ensure robust performance.

The results are presented in Tables 3 and 4, which compare the models in terms of accuracy, recall, precision and F-score. It should be noted that the results are not fully consistent across all metrics, which means that there is no single model that can be identified as the best one without ambiguity. From the perspective of accuracy, Random Forest demonstrates the highest performance, reaching 0.840 one year prior to distress and 0.906 two years prior. These values are considerably high and even exceed the results reported in the studies by Zhang et al. (2022) and Sreeharan et al. (2020). Interestingly, tree-based models such as Decision Tree, Random Forest and LightGBM show higher accuracy when predicting two years before financial distress compared to one year before the occurrence of financial distress. This suggests that such models are able to capture medium-term patterns in the data, which may be particularly useful for earlier identification of at-risk firms. Therefore, for medium-term predictions, tree-based models appear to be more suitable, with Random Forest being the most recommended one.

The next best performing model in terms of accuracy is LightGBM, which achieved 0.837 and 0.887 for one and two years prior to financial distress respectively. As expected, Logistic Regression had the lowest accuracy (0.657 and 0.592). This can be explained by the fact that Logistic Regression assumes a linear relationship between dependent and independent variables, whereas financial ratios can demonstrate non-linear patterns, that are better captured by non-linear models. Furthermore, after conducting the t-test/ Welch's test and multicollinearity checks, which led to the elimination of some features, only three variables remained in the Logistic Regression model. This may have limited the amount of available information, reducing its predictive ability and leading to lower accuracy scores.

Despite this, Logistic Regression achieved the highest recall scores across all models, equal to 0.704 and 0.923 for one and two years before financial distress models respectively. This means that the algorithm successfully identifies most of the companies in distress. However, this performance is accompanied by very low precision score, which indicates that among the companies predicted as distressed, a large proportion are in fact healthy. In general, the same pattern is observed for all models - recall scores are higher than precision. This indicates that models minimize Type II errors (false negatives) by not missing many distressed companies, but at the cost of producing a high number of false positives. On one hand, this could be considered a relatively safe outcome, as the models are unlikely to overlook firms in distress. On the other hand, it creates inefficiencies, since too many healthy companies are misclassified as distressed, leading to wasted time and resources in evaluating firms that do not actually face financial difficulties.

Overall, the Random Forest model can be considered the most balanced and effective. It achieves the highest accuracy and also the best F-score, which reflects a more reliable trade-off between recall and precision. This balance is particularly important for small and medium-sized enterprises (SMEs), which often operate under limited financial resources. For SMEs, a high number of false alarms can be costly, as additional analyses and investigations require time and money that could otherwise be invested in growth. At the same time, missing true cases of distress could have catastrophic consequences for such firms. Therefore, a model like Random Forest, which performs well in both accuracy and F-score, provides SMEs with a better balance between minimizing the risk of overlooking distress and avoiding excessive costs from false positives. Furthermore, its superior performance in the two-year-ahead prediction model is especially valuable, enabling managers to identify potential financial distress and implement corrective measures well in advance.

Table 3. Performance results for 1 year prior to financial distress model

Model	Accuracy	Recall	Precision	F-score
Logistic Regression	0.657	0.704	0.200	0.312
ANN	0.837	0.333	0.290	0.310
SVM	0.783	0.593	0.276	0.376
Decision Tree	0.796	0.444	0.255	0.324
Random Forest	0.840	0.444	0.333	0.380
LightGBM	0.837	0.297	0.276	0.286

Table 4. Performance results for 2 years prior to financial distress model

Model	Accuracy	Recall	Precision	F-score
Logistic Regression	0.592	0.923	0.108	0.192
ANN	0.829	0.308	0.108	0.160
SVM	0.771	0.462	0.109	0.175
Decision Tree	0.800	0.385	0.109	0.169
Random Forest	0.906	0.615	0.308	0.410
LightGBM	0.887	0.462	0.231	0.308

4.2 Explanation of the best model

After identifying the best-performing model, it is crucial to provide an interpretation of its results. Based on the performance evaluation, Random Forest was selected as the recommended model. Although it is built on decision trees, its ensemble nature makes it less transparent and harder to interpret, which may reduce transparency and stakeholder confidence in its application. As emphasised in the literature review, there is increasing recognition of the importance of interpretability to ensure that key stakeholders can both understand the model and apply its insights in practice. For this reason, a detailed explanation of the Random Forest model was provided, supported by explainable AI tools that generate both global and local interpretations of its predictions. Furthermore, while the model demonstrated stronger performance when predicting financial distress two

years in advance, explanations were developed for both one-year and two-year horizons to provide a more comprehensive understanding of the model's behaviour.

Before introducing the explainable AI tools, a broad overview of feature importance rankings was provided. Figures 2 and 3 illustrate the relative importance of variables for the one-year and two-year financial distress prediction models, respectively. In both the short-term and medium-term horizons, the current ratio emerges as the most influential predictor in distinguishing between healthy and distressed firms, followed by the debt-to-asset ratio. For managers of retail SMEs, these findings underscore the need to prioritise liquidity and leverage monitoring, treating these measures as key early-warning indicators. Establishing internal thresholds for the current ratio and debt-to-asset ratio could therefore provide a practical framework for assessing financial health and initiating timely corrective actions.

Overall, although two models produce relatively similar results, notable differences remain. For instance, in the one-year-before-distress model, profit margin plays a more significant role, whereas in the two-year model, cash flow-to-assets carries greater importance. This suggests that stakeholders should adjust their focus depending on the prediction horizon - in the short term, greater attention should be paid to profitability measures, while in the medium term, the cash flow indicator becomes a more critical factor of financial stability.

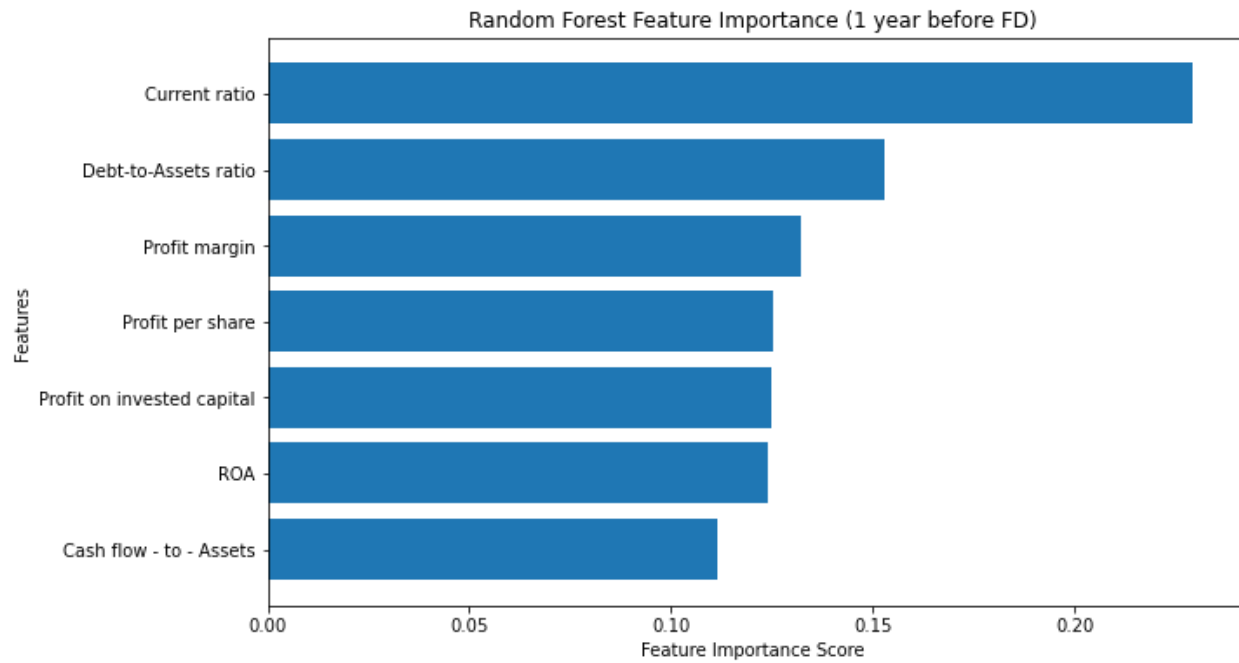


Fig.2. Feature importance for the one year before DF model

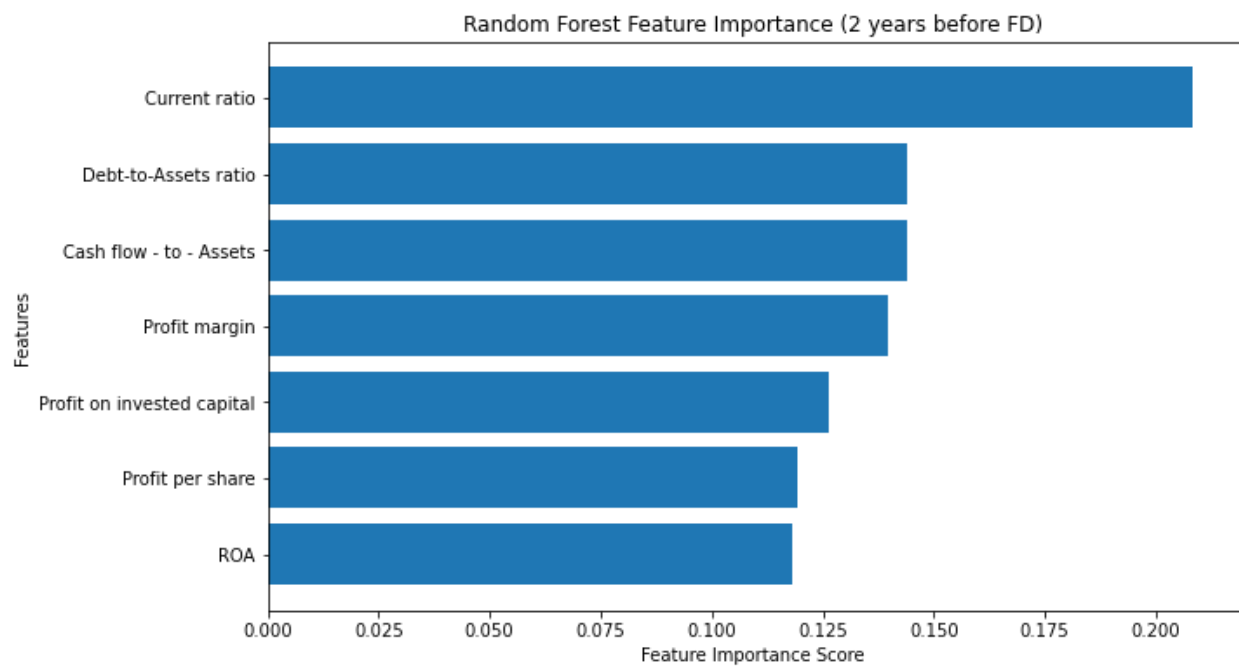


Fig.3. Feature importance for the two years before DF model

Although feature importance is a valuable tool, it does not provide insights into the direction of a variable's influence on the prediction. To obtain a more detailed understanding of the features' behaviour, an explainable AI approach was adopted through the use of TreeSHAP, implemented via the *shap* library. TreeSHAP is a specialised algorithm specifically designed for tree-based algorithms to efficiently calculate SHapley Additive exPlanations (SHAP) values. This method enables both global explanations, which highlight the overall impact of features across the dataset, and local explanations, which show how individual features contribute to predictions for specific observations.

One of the common global explanation tools for interpreting machine learning models is the beeswarm plot, shown in Figures 4 and 5. Similar to the feature importance rankings, these plots highlight the relative influence of variables on the prediction outcome. As illustrated, the current ratio remains the most influential factor, however, the overall ranking of features differs from that presented in Figures 2 and 3. These differences may arise because the rankings in Figures 2 and 3 are based on Gini importance, whereas Figures 4 and 5 reflect rankings calculated using SHAP values, which capture both the magnitude and direction of each variable's contribution to the model's predictions.

Beeswarm plots not only illustrate the importance of features but also provide insights into the direction of their influence. Each dot on the graph represents a company, with the colour of the dot reflecting the feature value - blue indicates a lower value, while red corresponds to a higher value. Dots positioned to the right of the central line push predictions towards financial distress, whereas those on the left move predictions towards a healthy classification. Overall, the influence of most variables aligns with general expectations. However, in both models an interesting pattern emerges - contrary to conventional outcome, a higher debt-to-assets ratio appears to decrease the probability of financial distress, as evidenced by the concentration of red dots (high values) on the left-hand side of the plot. This may be explained by the fact that retail SMEs, due to their reliance on inventory financing, thin margins, and exposure to consumer demand volatility, are especially dependent on access to external borrowing. In a sector, characterised by rapidly changing consumer preferences and volatile market conditions,

access to credit can provide essential flexibility to finance inventory, operations, and development. Consequently, maintaining a reasonable level of debt-to-assets ratio, rather than being overly conservative, may strengthen resilience. While this finding may seem counterintuitive in light of the current UK economic climate, marked by rising living costs and uncertainty, it highlights the importance of balanced leverage in ensuring financial stability for retail SMEs.

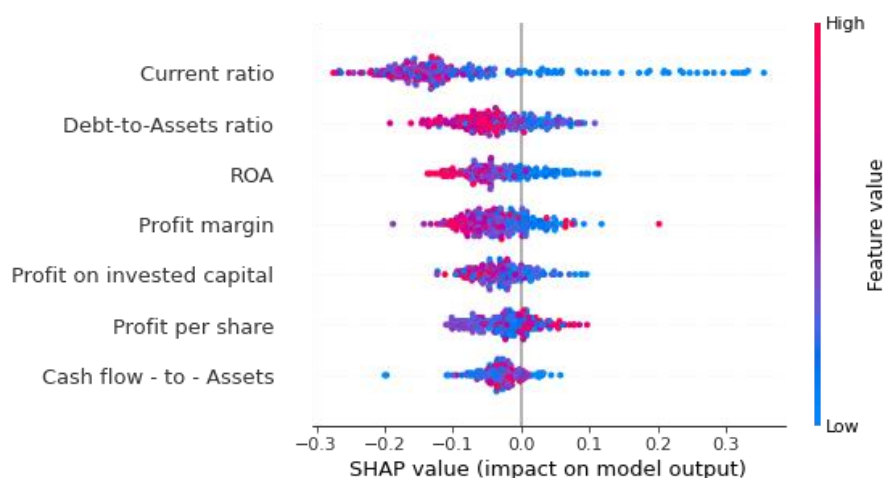


Fig.4. Beeswarm plot for one year prior to financial distress model

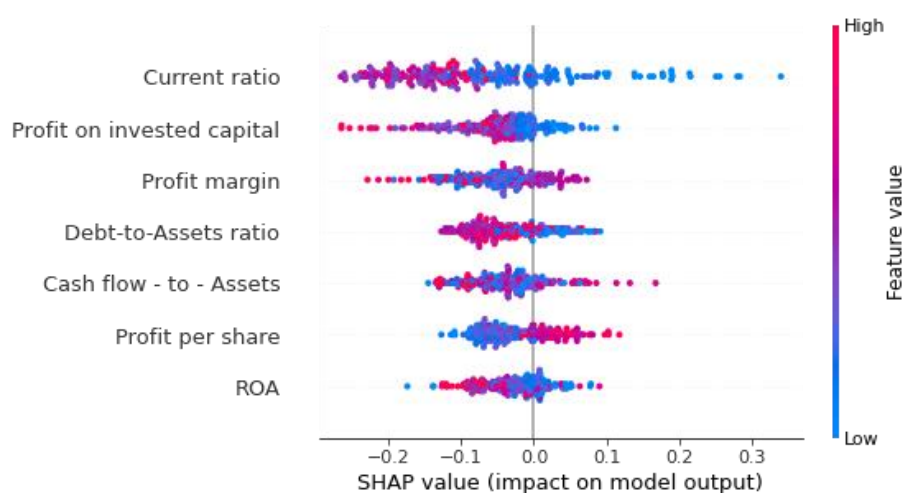


Fig.5. Beeswarm plot for two years prior to financial distress model

Another form of interpretability is provided by local explanations, which focus on individual predictions for a specific company. In this approach, the model output can be decomposed into three components: the baseline value, the contribution of each feature, and the final prediction. The baseline represents the average model prediction across all firms, while each financial variable shifts this baseline either upward (towards financial distress) or downward (towards financial health). It is important to note that the values shown next to each feature are not raw financial ratios but preprocessed values, indicating how far a company's feature lies above or below the average across the dataset. In this study, local explanations were generated for randomly selected distressed firms, one for the one year before distress model and another for the two years before distress model. These local explanations generated through force plots provide interesting insights into the key drivers of the predicted probability of financial distress.

Firstly, it is important to note that the base values in both figures are close to 0.5 - specifically, 0.4998 in Figure 6, and 0.4996 in Figure 7. This indicates that, before considering any financial features, the model assigns each firm an approximate 50% probability of financial distress. This outcome is logical given that the dataset was balanced using SMOTE, ensuring an equal representation of distressed and non-distressed firms.

As shown in Figure 6, which presents the force plot for the one year before distress model, the current ratio is the most significant factor pushing the prediction towards financial distress. This aligns with its overall ranking as the most important feature. The negative value (-0.9267) indicates that the company's current ratio is below the dataset average. In other words, insufficient liquidity of a company is driving the prediction upward. Another influential variable is the debt-to-assets ratio. Interestingly, its value is also below the average (-0.9149). The result appears counterintuitive as lower leverage is typically associated with reduced risk. This outcome may reflect a situation where the firm has limited access to credit markets and reduced financial flexibility, which in turn increases the likelihood of distress. Another noteworthy factor is profit per share, which for this company is above average (1.344), but still pushes the prediction upward. Although a higher Profit per Share (PPS) would traditionally be considered a positive signal of firm

performance, the model associates elevated PPS values with a higher risk of distress. Here, it is important to highlight that SHAP values must be understood in context, as the feature's influence is evaluated in combination. Thus, when paired with weak liquidity and solvency indicators, an elevated PPS may suggest that short-term shareholder returns are being prioritised at the expense of long-term financial stability. By contrast, features such as profit on invested capital, cash-flow-to-assets, and return on assets, which for this firm are all above average, contribute to lowering the probability of distress. Overall, however, the positive contributions are outweighed by the negative ones, resulting in a final model prediction of 0.56, classifying the firm as financially distressed.

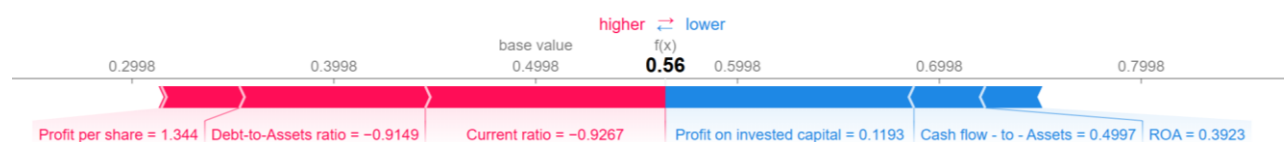


Fig.6. Force Plot for one year prior to financial distress model

Figure 7 presents the feature contributions for the two years before distress model. Overall, the results are broadly consistent with the one-year model, though the relative strength of the feature influences differs. For example, Profit per Share shows a stronger impact than the debt-to-assets ratio in this case. What is interesting, is the influence of current ratio variable. Although the company's current ratio is also below the dataset average, it nonetheless contributes to lowering the probability of distress, in contrast to the previous case. This suggests the possibility that a certain threshold exists, where only more substantial negative deviations from the average become critical in pushing the prediction upward. In cases where the current ratio is modestly below average, the model may interpret it as less concerning, thereby pulling the prediction downwards. This is illustrated in Figure 7, where the final model prediction is 0.36, indicating that the firm is classified as financially healthy despite some weaker liquidity signals.

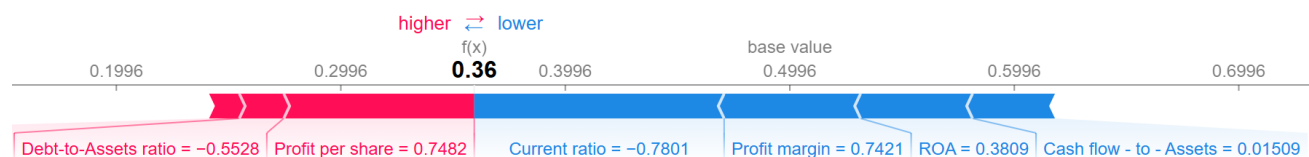


Fig.7. Force Plot for two years prior to financial distress model

Overall, the model produces results, that sometimes can seem counterintuitive (such as direction of the influence of the debt-to-assets ratio on financial distress), highlighting the value of using explainable AI tools. Thus, a key advantage of SHAP is that it does not assess each feature in isolation but rather evaluates how features interact in combination. As a result, the typical influence of a given variable may shift when considered alongside other financial indicators, occasionally producing effects that appear to contradict common sense. This nuance is particularly important for stakeholders, as it encourages them to interpret financial signals holistically rather than focusing on single ratios. By understanding how different factors reinforce or offset one another, managers and investors can make more informed and tailored decisions that better reflect the complex realities of financial distress prediction.

5. Conclusions

The current study focused on identifying the most accurate machine learning model that can accurately distinguish between financially distressed and healthy companies in the UK SME retail companies.

It is important to note that the research process was iterative, with some objectives and methods refined throughout the literature review, analysis, and modelling stages in order to better align with the overall aim. The study's aim was shaped by an examination of the current business environment in the UK retail SME sector, together with insights from the literature, which underscored both the relevance of the topic and the existing research gap. Although the number of studies on financial distress prediction has been steadily increasing, to the best of our knowledge there are no recent works focusing specifically on UK retail SMEs. This gap is particularly significant in light of recent reports from insolvency experts, which highlight a sharp rise in the number of UK firms facing "critical" financial distress. At the same time, SMEs are central to the UK economy, accounting for sufficient part of the business population at the start of 2024 and acting as a foundation for local communities through job creation and economic contribution.

After the analysis of the literature review, six models were selected for comparison - Logistic Regression, ANN, Support Vector Machine SVM, Decision Tree, Random Forest and LightGBM. Logistic Regression was chosen as the benchmark model. The study also aimed to compare the performance of single classifiers with ensemble classifiers in the context of UK retail SMEs. Following a rigorous data preparation process, outlined in the data workflow section, model evaluation was carried out to identify the most effective approach using standard metrics, including accuracy, precision, recall and F-score. To optimise performance and reduce the risk of overfitting, hyperparameter tuning was applied alongside 5-fold cross-validation.

Based on the results of the evaluation metrics, the Random Forest model emerges as the most effective and balanced approach, demonstrating the strongest performance in both accuracy and F-score. The high F-score indicates that it manages the trade-off between precision and recall more reliably than other models. This balance is especially valuable for small and medium-sized enterprises (SMEs), which often face constraints in financial and operational resources. Excessive false alarms can drain these resources, diverting attention and funds away from growth opportunities, while failing to detect genuine distress cases may result in severe consequences. By achieving strong results across both metrics, Random Forest offers SMEs a practical solution that reduces the likelihood of overlooking financial distress while also limiting unnecessary costs from false positives.

Since Random Forest is not inherently transparent, additional explanation was necessary to provide reliability and applicability for key stakeholders. To begin, feature importance rankings were examined, showing that the current ratio and debt-to-asset ratio were consistently the most influential variables in both the one year before financial distress and two years before financial distress prediction models. To provide deeper insights, the explainable AI tool TreeSHAP was applied, offering both global and local interpretations of the model. The global explanations, illustrated by beeswarm plots, revealed an interesting pattern - a higher debt-to-asset ratio was associated with a lower probability of financial distress. This counterintuitive finding may reflect the fact that easy access to credit markets can give retail SMEs greater financial flexibility, enabling them to adapt to constantly changing business conditions. Local explanations provided through force plots

offered even more nuanced insights, showing that the effect of the current ratio on the model's prediction varies depending on the degree of its deviation from the average and its interaction with other variables. This highlights one of the main strengths of SHAP values, that account for combinations of features at the firm level rather than treating them in isolation.

Overall, local explanations are particularly valuable because they reveal why the model makes a specific prediction for a given company. This improves transparency and allows firms to design targeted responses. Such tailored decision-making is especially important for SMEs, which typically operate with limited resources. By filtering for companies with similar profiles, managers can better understand how financial factors interact in their specific context, enabling more accurate, data-driven decisions. This not only helps SMEs avoid financial distress but also reduces unnecessary costs, thereby supporting their long-term development.

Overall, it can be concluded that the aims and objectives of this study have been achieved. Nevertheless, this study has several limitations that can motivate further research. The first limitation is connected with the fact that the data is trained on the UK SME's specialising on retail companies' data from 2020 to 2024. This may limit the generalisability of the findings, as the results may not be directly applicable to firms in other industries or to larger enterprises. In addition, the models may require retraining on updated datasets in the event of major future economic shocks, to ensure that new market conditions are properly incorporated. A second limitation is that the study relies on a limited set of accounting-based variables. Future research could extend this by incorporating a broader range of predictors, including macroeconomic indicators and corporate social responsibility (CSR) measures. Expanding both the number of variables and the dataset size would likely improve accuracy and produce better results in terms of F-score. In addition, integrating accounting-based indicators with legal records in order to flag financially distressed companies could create a classification framework that more accurately reflects real-world conditions. Finally, the use of hybrid models that integrate the strengths of different classifiers may help overcome the limitations of individual algorithms and further enhance predictive performance.

6. References

1. Abrahamsen, N. G. B., Nylén-Forthun, E., Møller, M., de Lange, P. E., & Risstad, M. (2024). Financial Distress Prediction in the Nordics: Early Warnings from Machine Learning Models. *Journal of Risk and Financial Management*, 17(10), Article 432. <https://doi.org/10.3390/jrfm17100432>
2. Aydin N., Sahin N., Deveci M., & Pamucar D. (2022). Prediction of financial distress of companies with artificial neural networks and decision trees models. *Machine Learning with Applications*, 10, 100432. <https://doi.org/10.1016/j.mlwa.2022.100432>
3. Begbies Traynor Group. (n.d.). Historic jump in the number of firms in critical financial distress. <https://www.begbies-traynorgroup.com/news/business-health-statistics/historic-jump-in-the-number-of-firms-in-critical-financial-distress>
4. Colff, F., & Brümmer, L. (2019). Financial distress prediction using a machine learning model: A study of JSE-listed companies. *Management Dynamics*, 28(4), 29–40.
5. FasterCapital. (2025, April 05). Cash flow to asset ratio: how to calculate and interpret cash flow to asset ratio. <https://fastercapital.com/content/Cash-flow-to-assets-ratio--How-to-Calculate-and-Interpret-Cash-Flow-to-Assets-Ratio.html>
6. Fernando, J. (2025, May 17). Current ratio explained with formula and examples. Investopedia. <https://www.investopedia.com/terms/c/currentratio.asp>
7. GeeksforGeeks. (2024, May 3). SMOTE for Imbalanced Classification with Python. <https://www.geeksforgeeks.org/machine-learning/smote-for-imbalanced-classification-with-python/>
8. GeeksforGeeks. (2025, July 21). ML | Handling Missing Values. <https://www.geeksforgeeks.org/machine-learning/ml-handling-missing-values/>
9. Government Commercial Function. (2023). Supplementary information: Small and Medium-sized Enterprises definition. https://assets.publishing.service.gov.uk/media/6731e5cff8ac0a8bd93d138a/Supplementary_Guidance_-_Defining_Small_and_Medium_Sized_Enterprises_FINAL.pdf
10. Gruszczyński, M. (2019). Modeling Financial Distress and Bankruptcy. In *Financial Microeconometrics* (pp. 77–119). Springer International Publishing AG. https://doi.org/10.1007/978-3-030-34219-7_3

11. Htoon, K. (2020, May 11). Levene's Test: The Assessment for Equality of Variances. Medium. <https://medium.com/@kyawsawhtoon/levenes-test-the-assessment-for-equality-of-variances-94503b695a57>
12. Jack, S. & Edwards, C. (2025, January 24). 'Costs just keep rising' - jump in firms in trouble. BBC News. <https://www.bbc.co.uk/news/articles/c9vmrpdrk4eo>
13. Kartika, G. (2020, October 13). Handling missing values. Saltfarmer's Blog. <https://saltfarmer.github.io/blog/python/Handling-Missing-Value/>
14. Koli, S. (2023, February 27). Decision Trees: A Complete Introduction With Examples. Medium. <https://medium.com/@MrBam44/decision-trees-91f61a42c724>
15. Liang, D., Tsai, C.-F., & Wu, H.-T. (2015). The effect of feature selection on financial distress prediction. Knowledge-Based Systems, 73(1), 289–297. <https://doi.org/10.1016/j.knosys.2014.10.010>
16. Martono, N. P., & Ohwada, H. (2023). Financial distress model prediction using Machine Learning: A case study on Indonesia's Consumers Cyclical companies. Machine Learning and Principles and Practice of Knowledge Discovery in Databases, 1753, 53–61. https://doi.org/10.1007/978-3-031-23633-4_5
17. Mselmi, N., Lahiani, A., & Hamza, T. (2017). Financial distress prediction: The case of French small and medium-sized firms. International Review of Financial Analysis, 50, 67–80. <https://doi.org/10.1016/j.irfa.2017.02.004>
18. Parker, A. (2023, August 29). SME companies in UK – A complete guide. London Local News. <https://www.londonlocalnews.co.uk/sme-companies-in-uk/>
19. Segal, T. (2025, May 17). Profit Margin: Definition, types, uses in business and investing. Investopedia. <https://www.investopedia.com/terms/p/profitmargin.asp>
20. Sehgal, S., Mishra, R. K., Deisting, F., & Vashisht, R. (2021). On the determinants and prediction of corporate financial distress in India. Managerial Finance, 47(10), 1428–1447. <https://doi.org/10.1108/MF-06-2020-0332>
21. Siddiqui, K. (2019, May 23). How Multicollinearity effects your model? Medium. <https://medium.com/@oraware/how-multicollinearity-effects-your-model-d68eaec7e7fb>
22. Skilling Writers Team. (2024, July 25). ROA (Return on Assets) meaning. <https://skilling.com/eu/en/blog/trading-terms/roa/>

23. Smith, A. (2024, September 30). What are the advantages and disadvantages of random forest? Pickl.AI. <https://www.pickl.ai/blog/advantages-and-disadvantages-random-forest/>
24. Song, Y., Jiang, M., Li, S., & Zhao, S. (2024). Class-imbalanced financial distress prediction with machine learning: Incorporating financial, management, textual, and social responsibility features into index system. *Journal of Forecasting*, 43(3), 593–614. <https://doi.org/10.1002/for.3050>
25. Sreedharan, M., Khedr, A. M., & El Bannany, M. (2020). A Comparative Analysis of Machine Learning Classifiers and Ensemble Techniques in Financial Distress Prediction. 2020 17th International Multi-Conference on Systems, Signals & Devices (SSD), 653–657. <https://doi.org/10.1109/SSD49366.2020.9364178>
26. Sugavasilakshmisahithi. (2025, February 23). Winsorization: a simple and effective way to handle outliers in your data. Medium. <https://medium.com/@sugavasilakshmisahithi/winsorization-a-simple-and-effective-way-to-handle-outliers-in-your-data-e9635c3bc049>
27. The Perfect Answers. (2024, December 7). How to Figure Out Profit Margin: Step-by-Step. Medium. <https://medium.com/@malokaadvocate/how-to-figure-out-profit-margin-step-by-step-c943e0b7d1df>
28. Tinoco, M., & Wilson, N. (2013). Financial distress and bankruptcy prediction among listed companies using accounting, market and macroeconomic variables. *International Review of Financial Analysis*, 30, 394–419. <https://doi.org/10.1016/j.irfa.2013.02.013>
29. Ward, M. (2025, April 25). Retail sector in the UK. Houses of Commons Library. <https://researchbriefings.files.parliament.uk/documents/SN06186/SN06186.pdf>
30. Zizi, Y., Jamali-Alaoui, A., El Goumi, B., Oudgou, M., & El Moudden, A. (2021). An optimal model of financial distress prediction: A comparative study between neural networks and logistic regression. *Risks (Basel)*, 9(11), Article 200. <https://doi.org/10.3390/risks9110200>

Appendix 1. Additional materials

Link to OneDrive: https://universityofexeteruk-my.sharepoint.com/:f:/r/personal/vt326_exeter_ac_uk/Documents/BEMM466_Titova?csf=1&web=1&e=2CpZf6

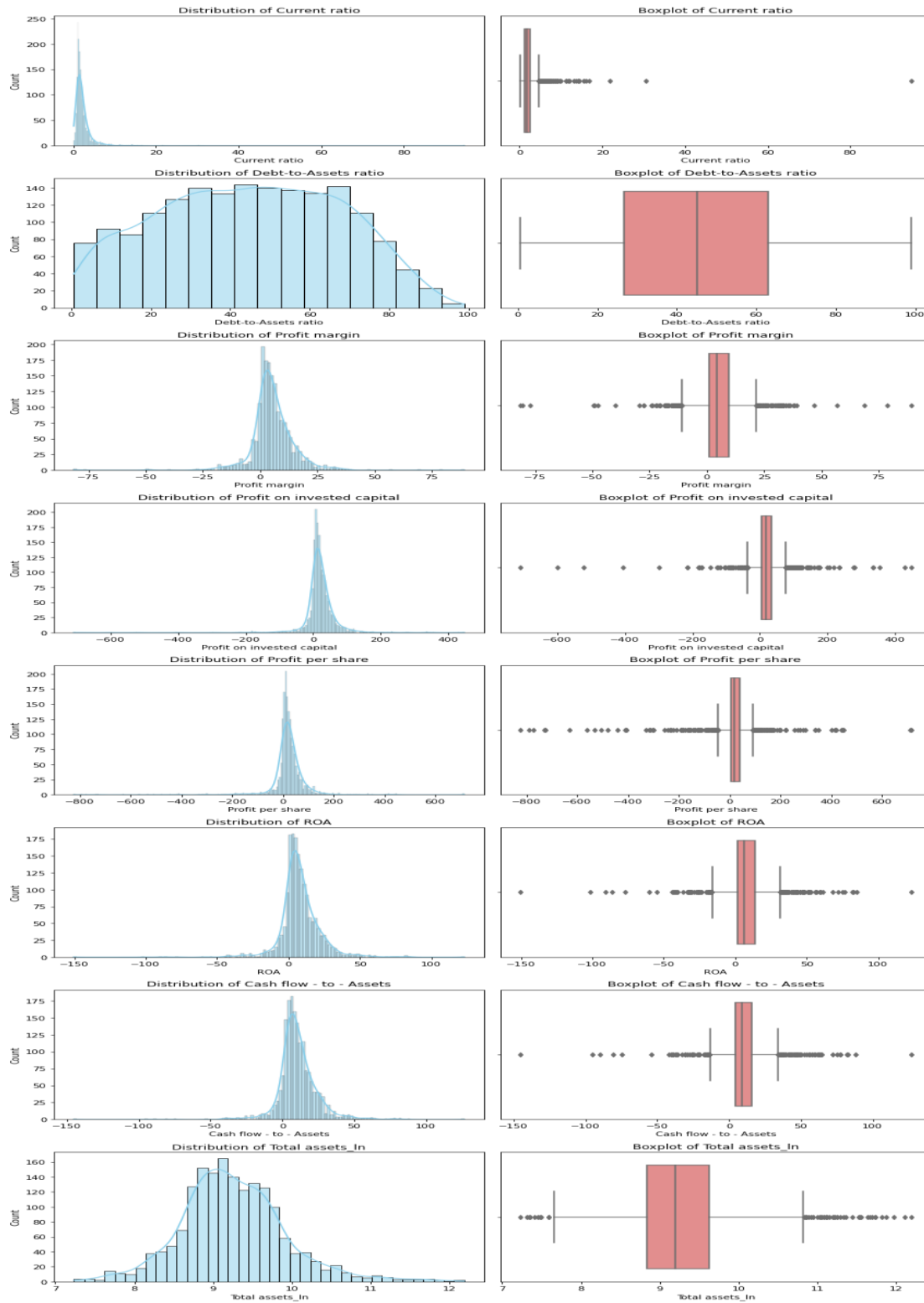


Fig.8. Distributions and box plot of the data before the outliers

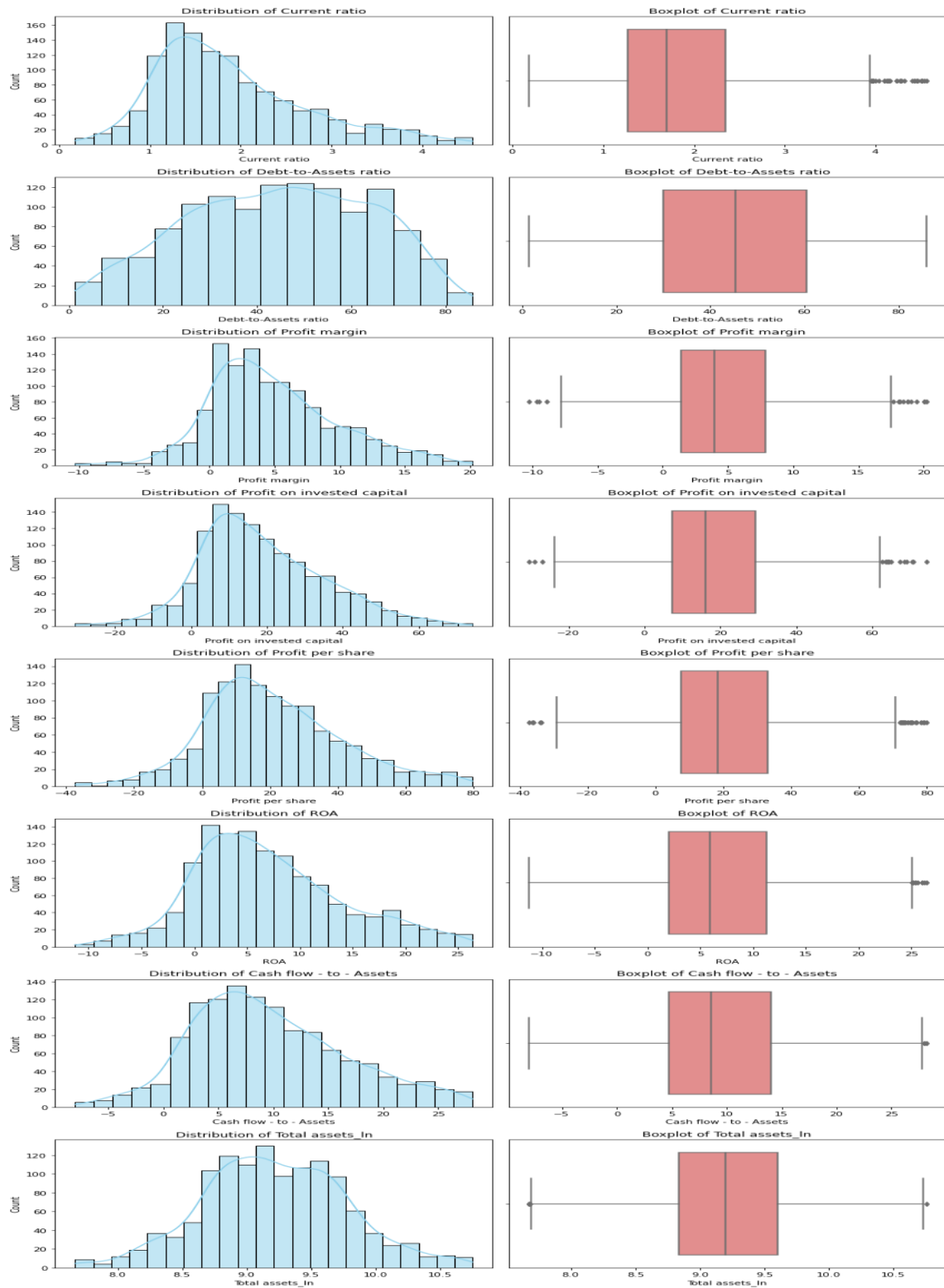


Fig. 9. Distributions and box-plots after data cleaning

	Variable	Test type	p-value	Result
0	Current ratio	Standard t-test	0.00	Significant difference
1	Debt-to-Assets ratio	Welch's t-test	0.00	Significant difference
2	Profit margin	Standard t-test	0.00	Significant difference
3	Profit on invested capital	Standard t-test	0.00	Significant difference
4	Profit per share	Welch's t-test	0.00	Significant difference
5	ROA	Welch's t-test	0.00	Significant difference
6	Cash flow - to - Assets	Standard t-test	0.00	Significant difference
7	Total assets_In	Standard t-test	0.33	No significant difference

Fig. 10. Results of T-test and Welch's test

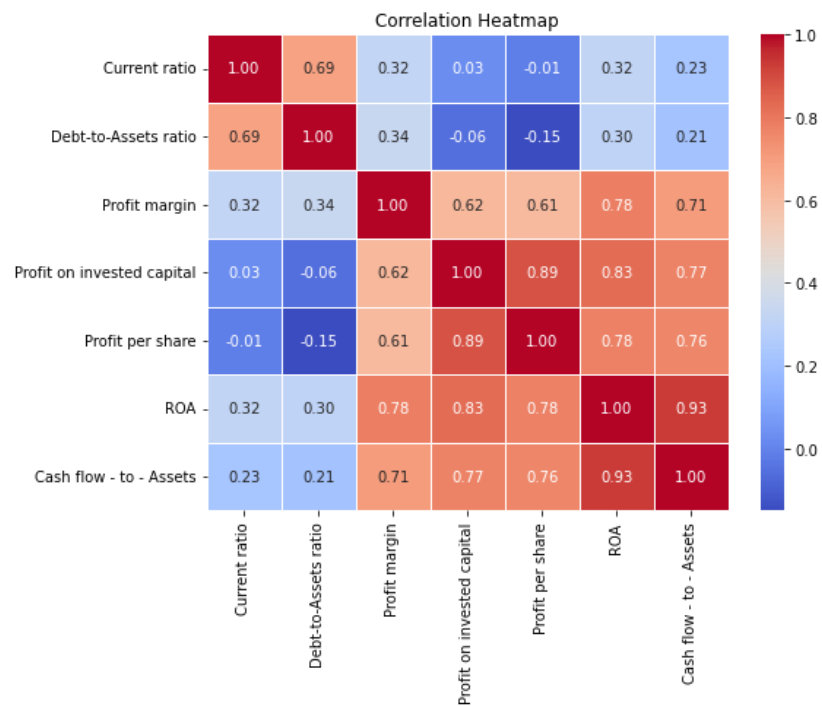


Fig. 11. Correlation matrix with all the independents variables

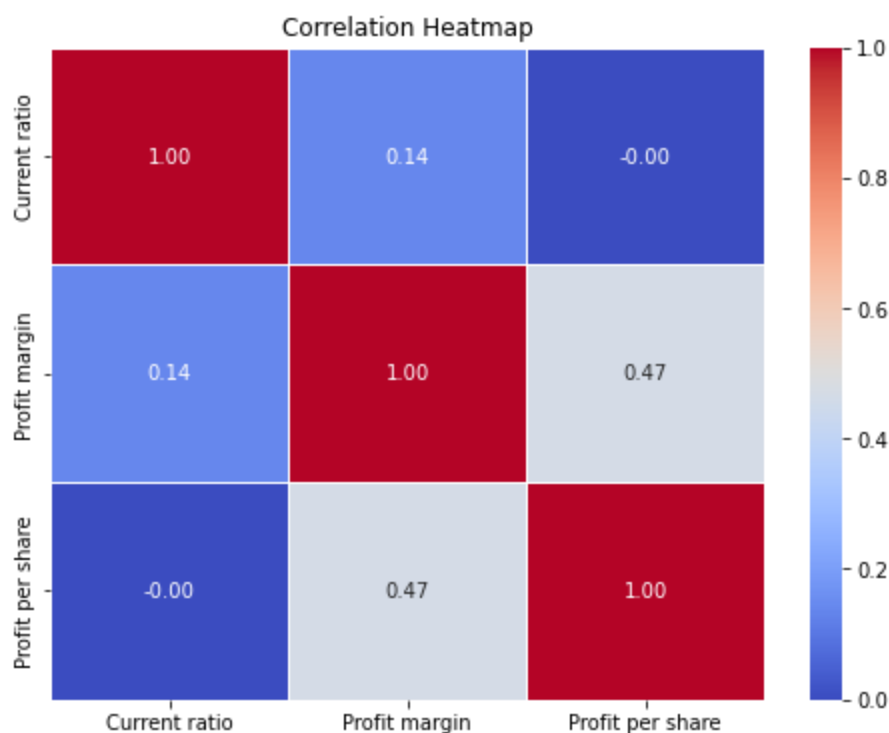


Fig. 12. Correlation matrix after removing highly correlated variables

Year	Distressed	Count	Percentage
2020	0	175	83.732057
2020	1	34	16.267943
2021	0	211	86.475410
2021	1	33	13.524590
2022	0	273	88.064516
2022	1	37	11.935484
2023	0	293	87.462687
2023	1	42	12.537313
2024	0	112	88.188976
2024	1	15	11.811024

Fig. 13. Percentage of companies flagged as distressed and non-distressed, grouped by year

Appendix 2. Project proposal

1. Introduction

The topic of predicting company financial distress has long attracted the interest of academics and practitioners. However, developing a universal model is challenging due to the evolving nature of business environment and the presence of various national and regional factors that affect firms operating activity, which can drive ongoing research into more context-sensitive approaches.

It is worth to mention that the development of the accurate machine learning model would be highly relevant in the context of the UK. Thus, recent reports indicate a sharp rise in businesses facing “critical” financial distress - cases surged by 50.2% from 31,201 to 46,853 between Q3 and Q4 of 2024 (Jack & Edwards, 2025). Consumer-facing sectors, such as retail, hospitality, and leisure, were among the most affected (Begbies Traynor Group, n.d.). While firm closures are a natural part of economic cycles, atypical exit of companies driven by financial distress can have severe negative effects on the broader economy (Sehgal et al., 2021). For instance, the retail sector alone generated £111.6 billion in output in 2024, representing 4.5% of the UK’s total output (Ward, 2025). Thus, improving early detection of financial distress in this sector could help mitigate significant economic losses.

The aim of this research is to develop a machine learning model that can accurately distinguish between financially distressed and healthy companies in the UK SME retail companies. To support this aim, the following objectives and questions have been identified that are presented in Table 1.

Research objective	Research questions
Identify the most relevant features of financial distress	Which indicators have been most commonly used in prior research to predict corporate financial distress? Are there notable differences in the average values of identified indicators between distressed and non-distressed companies?

	What are the most significant features of financial distress?
Determine the optimal prediction time horizon	What are the most commonly used time horizons for financial distress prediction in existing literature? Which time horizon provides better model performance based on evaluation metrics?
Identify the best machine learning model	Which machine learning models are most suited to achieve the aim of current study? Which machine learning models have shown the strongest performance based on evaluation metrics?

Table 1. Identified research objectives and corresponding questions.

The outcomes of this project will be valuable to a range of stakeholders, including managers, investors, and creditors (Mselmi et al., 2017; Sreedharan et al., 2020). The decision to focus on SME's is motivated by the fact that they play a vital role in the UK economy, representing 99.9% of the business population at the start of 2024 and acting as a foundation for local communities through job creation and economic contribution (Parker, 2023; Ward, 2025). Having access to an accurate prediction model can help SME owners take preventive action. Therefore, a reliable model for predicting SME financial distress would deliver widespread economic and social benefits.

2. Literature review

2. 1. Identifying target variable

The initial step in modelling financial distress is to clearly define the concept in order to determine which companies can be classified as “distressed”. A key challenge in this process is the broad nature of the term, which has led to the absence of a universally accepted definition. Broadly, financial distress refers to a company's inability to generate sufficient operating cash flow to meet its financial obligations, often requiring corrective action (Gruszczynski, 2020).

It is worth to mention that different studies adopt various criteria to classify financial distress. Some researchers use bankruptcy status as a proxy, while others rely on legal or institutional definitions (Aydin et al., 2022; Tinoco & Wilson, 2013). For instance, Mselmi et al. (2017) define distress based on judicial declarations, whereas Zizi et al. (2021) consider firms as distressed if they fail to meet credit obligations for over 90 days. However, in order to facilitate early detection, many studies employ financial indicators to identify distress before formal insolvency occurs (Abrahamsen et al., 2024; Martono & Ohwada, 2023; Sreedharan et al., 2020; Tinoco & Wilson, 2013; Sanjay & Mishra, 2020). For instance, Abrahamsen et al. (2024) classify firms as distressed when the interest coverage ratio is below 1.5 and the current ratio is under 1, while Martono and Ohwada (2023) use negative operating cash flow. Tinoco and Wilson (2013) propose a hybrid model combining three criteria: (1) EBITDA below financial expenses for two consecutive years, (2) a two-year decline in market value, and (3) formal default—merging legal, market-based, and accounting-based definitions to improve classification accuracy.

2.2. Independent variables

Identifying key predictors of distress is the next important step of model construction. Beaver's early research showed financial ratios have strong explanatory power. Building on that, Altman developed multivariate models confirming the predictive value of financial ratios (Mselmi et al., 2017; Sehgal et al., 2020). Consequently, financial ratios remain widely used in recent research. However, as businesses and data environments become more complex, many scholars note that financial ratios, that are based on historical, quantitative data, may have limited predictive power as they may contain an outdated information (Song et al., 2022; Colff & Brummer, 2019; Tinoco & Wilson, 2013). Recent studies advocate combining financial variables with non-financial indicators, such as management quality, social responsibility factors, and macroeconomic variables (Song et al., 2022; Colff & Brummer, 2019; Judijanto et al., 2024). For example, Song et al. (2022) in order to analyse the influence of management structure include such indicators as number of directors, number of independent directors, number of shares held by executives and whether the roles of chairman and CEO are separated.

Once variables are selected, it is important to eliminate redundant features to prevent overfitting, reduce dimensionality, and improve model accuracy (Zizi et al., 2021; Liang et al., 2015). What is more, feature selection can also decrease model training time and improve its generalization ability (Song et al., 2024). Feature selection methods generally fall into two categories: filter-based and wrapper-based. Liang (2015) compared several methods and found no universal best approach since performance varied depending on the dataset. Thus, Genetic Algorithms (wrapper-based) performed well on credit scoring data, while Logistic Regression (filter-based) was more effective for bankruptcy prediction. Other examples include Stepwise and Lasso Logistic Regression (Zizi et al., 2021), TreeSHAP (Abrahamsen et al., 2024), and Backward Elimination (Mselmi et al., 2017). Ultimately, the choice of feature selection method can depend on the classification task and computational resources available.

2.3. Models

When it comes to the modelling step, financial distress prediction methods broadly fall into statistical and machine learning (ML) models. Statistical models are considered as easy to use and interpret, but rely on assumptions such as linearity, normality, and variable independence, which may not hold in real-world data (Song et al., 2023; Colff & Brummer, 2019). Due to increasing data complexity, many recent studies apply machine learning methods, which can offer higher accuracy, flexibility, and the ability to process complex, multidimensional data (Zizi, 2021; Sreedharan et al., 2020).

ML models can be broadly categorized into single and ensemble classifiers. Single classifiers such as Decision Trees (DT), Support Vector Machines (SVM), and Artificial Neural Networks (ANN) are widely used for their speed and performance. At the same time, ensemble classifiers including voting-based, bagging-based, and boosting-based models are gaining more attention for their superior predictive capabilities (Song et al., 2023). For instance, Martono and Ohwada (2023) compare the performance of 4 models – Logistic Regression (LR), Decision Tree (DT), LightGBM and KNN. Empirical result demonstrates the accuracy range between 0.60 and 0.87, where the highest accuracy value of 0.87 was achieved by LightGBM algorithm for the scenario of 1 year prior to the financial distress. This aligns with the result of Zhang et al. (2022), where LightGBM

model shows the best performance across all the presented metrics. However, ensemble models do not always perform best. Sreedharan et al. (2020) highlight that despite the fact that Majority Voting classifier showed the highest average accuracy and outperform the results of AdaBoost and RF, it could not outperform the average accuracy of Neural Network classifier. Similarly, comparison between statistical and machine learning also show contradictory outcomes. Thus, Zizi et al. (2021) found that Logistic Regression outperformed ANN, with accuracy of 95% versus 88.33%. Therefore, no single classifier consistently performs best and the result might be highly dependent on dataset characteristics.

2.4. Discussion on explainable AI

As mentioned earlier, machine learning models are increasingly popular for financial distress prediction due to their ability to handle complex, non-linear relationships and their flexibility compared to traditional statistical models, which often rely on restrictive and unrealistic assumptions. However, some ML models function as “black boxes”, offering limited transparency and interpretability. This lack of clarity can lead to a reduced trust from the key stakeholders and hinder decision-making. As a result, there is growing concern about the importance of providing model interpretation to facilitate stakeholders’ understanding and confidence (Judijanto et al., 2024; Zhang et al., 2022). To address this issue, explainable AI (XAI) tools have been developed to improve model transparency. Common methods include SHapley Additive exPlanations (SHAP), Partial Dependence Plots (PDP), and counterfactual explanations, which help to provide global and local explanations and give some practical insights to the stakeholders (Zhang et al., 2022; Liu et al., 2022).

In summary, it can be seen that the topic of financial distress is really comprehensive and includes different aspects to analyse such as identifying distressed companies, selecting significant features and models and providing the interpretation of result. Moreover, the model performance is highly context-dependent, which motivates to farther exploration of this topic and finding the best-performing model on specific data. Thus, For UK retail SMEs that face unique risk profiles and economic pressures customized, interpretable, and robust ML models are essential.

3. Modelling

3.1 Data

To address the research aims and objectives, secondary data will be sourced from the Orbis database by Bureau van Dijk. Orbis is a reputable source of data that integrates information on over 580 million companies globally, drawing from official sources such as company registries, credit agencies, and regulators. Its rigorous data standards ensure high reliability, supporting the relevance and applicability of this study's findings to real-world business scenarios.

3.2 Target variable

This research focuses on small and medium-sized enterprises (SMEs) in the UK retail sector. According to UK government criteria, a company qualifies as an SME if it has fewer than 250 employees and either a turnover of $\leq$ £44 million or a balance sheet total of $\leq$ £38 million (Government Commercial Function, 2023). These criteria, along with the retail sector filter, will be applied in Orbis to extract data for the target group.

3.3. Data preprocessing

After acquiring the raw data, several data preparation steps will be applied to ensure proper model functionality. First, missing values will be addressed, as the selected models cannot handle them directly. Generally, missing data can be managed through imputation or removal, the former is suitable when the proportion of missing values is low. However, since the focus of this study is on SMEs, a sufficient number of missing values is expected. Therefore, observations with missing values will be removed, and features with a large proportion of missing data will be dropped. Additionally, some models, such as Logistic Regression and SVM, are sensitive to outliers. To detect and treat outliers, the winsorisation technique will be used, as it is simple, preserves data, and yields interpretable results (Sugavasalakshmisahithi, 2025). The dataset is also expected to be imbalanced, with distressed companies underrepresented, which may bias results and increase Type I errors. To address this, SMOTE (Synthetic Minority Over-sampling Technique) will be applied. Finally, to reduce the risk of overfitting, a feature selection process will be implemented to retain only relevant predictors.

3.4 Chosen machine learning models

In this study it is intended to compare the following models – Logistic Regression, ANN, SVM, Decision Tree, Random Forest and LightGBM. Logistic Regression was chosen as a benchmark model due to its simplicity and interpretability. However, LR requires the linear relationship between target and independent variables, is sensitive to outliers, which may limit its performance (Mselmi et al., 2017; GeeksforGeeks, 2025). Therefore, ML models are also included.

As to single classifiers, it is intended to use Artificial Neural Network, Support Vector Machine and Decision Trees. DT was selected for its low complexity and high interpretability. ANN and SVM, though considered “black box” models, they don’t rely on strict assumptions and are able to capture linear and non-linear patterns in data. The disadvantage of these models is that they offer low interpretability as well as there is difficulty fixing the parameters of the model, which requires hyperparameter tuning.

As to ensemble classifiers, RF and LightGBM were chosen for the analysis and represent bagging and boosting approaches, respectively. RF is widely used due to its ability to output feature importance and its robustness to overfitting (Smith, 2024). LightGBM was selected over GBDT and XGBoost as it addresses their limitations—it is faster, uses less memory, and is less computationally intensive while maintaining high performance (Qian et al., 2022). The hyperparameter tuning will be applied for such models as SVM, ANN, DT, RF and LightGBM. Moreover, in order to avoid overfitting – cross- validation technique will be applied.

3.5. Results

After estimating the models, their performance will be evaluated and compared using accuracy, precision, recall, and F-score, and the most effective model will be selected for final interpretation. If the best performing model would be “black-box” model, the explainable AI methods such as SHAP (or TreeSHAP for tree-based models) are going to be used in order to make the model interpretable.

4. Limitations

Before applying the results of this research in practice, it is important to acknowledge several limitations. First, the model is trained on data from 2020 to 2024, which may limit its generalisability, especially in the event of major future economic shocks. Second, this study focuses exclusively on accounting-based variables due to scope constraints. Future research could enhance predictive performance by incorporating additional features, such as textual data or macroeconomic indicators. Lastly, the use of hybrid models may help address the limitations of individual classifiers and further improve prediction accuracy.

5. References

Abrahamsen, N. G. B., Nylén-Forthun, E., Møller, M., de Lange, P. E., & Risstad, M. (2024). Financial Distress Prediction in the Nordics: Early Warnings from Machine Learning Models. *Journal of Risk and Financial Management*, 17(10), Article 432. <https://doi.org/10.3390/jrfm17100432>

Aydin N., Sahin N., Deveci M., & Pamucar D. (2022). Prediction of financial distress of companies with artificial neural networks and decision trees models. *Machine Learning with Applications*, 10, 100432. <https://doi.org/10.1016/j.mlwa.2022.100432>

Begbies Traynor Group. (n.d.). Historic jump in the number of firms in critical financial distress. <https://www.begbies-traynorgroup.com/news/business-health-statistics/historic-jump-in-the-number-of-firms-in-critical-financial-distress>

GeeksforGeeks. (2025, August 2). Logistic regression in machine learning. <https://www.geeksforgeeks.org/machine-learning/understanding-logistic-regression/>

Government Commercial Function. (2023). Supplementary information: Small and Medium-sized Enterprises definition. https://assets.publishing.service.gov.uk/media/6731e5cff8ac0a8bd93d138a/Supplementary_Guidance_-_Defining_Small_and_Medium_Sized_Enterprises_FINAL.pdf

Gruszczyński, M. (2019). Modeling Financial Distress and Bankruptcy. In *Financial Microeconometrics* (pp. 77–119). Springer International Publishing AG. https://doi.org/10.1007/978-3-030-34219-7_3

- Tinoco, M., & Wilson, N. (2013). Financial distress and bankruptcy prediction among listed companies using accounting, market and macroeconomic variables. *International Review of Financial Analysis*, 30, 394–419. <https://doi.org/10.1016/j.irfa.2013.02.013>
- Jack, S. & Edwards, C. (2025, January 24). 'Costs just keep rising' - jump in firms in trouble. BBC News. <https://www.bbc.co.uk/news/articles/c9vmrpdrk4eo>
- Liang, D., Tsai, C.-F., & Wu, H.-T. (2015). The effect of feature selection on financial distress prediction. *Knowledge-Based Systems*, 73(1), 289–297. <https://doi.org/10.1016/j.knosys.2014.10.010>
- Martono, N. P., & Ohwada, H. (2023). Financial distress model prediction using Machine Learning: A case study on Indonesia's Consumers Cyclical companies. *Machine Learning and Principles and Practice of Knowledge Discovery in Databases*, 1753, 53–61. https://doi.org/10.1007/978-3-031-23633-4_5
- Mselmi, N., Lahiani, A., & Hamza, T. (2017). Financial distress prediction: The case of French small and medium-sized firms. *International Review of Financial Analysis*, 50, 67–80. <https://doi.org/10.1016/j.irfa.2017.02.004>
- Parker, A. (2023, August 29). SME companies in UK – A complete guide. London Local News. <https://www.londonlocalnews.co.uk/sme-companies-in-uk/>
- Sehgal, S., Mishra, R. K., Deisting, F., & Vashisht, R. (2021). On the determinants and prediction of corporate financial distress in India. *Managerial Finance*, 47(10), 1428–1447. <https://doi.org/10.1108/MF-06-2020-0332>
- Smith, A. (2024, September 30). What are the advantages and disadvantages of random forest? Pickl.AI. <https://www.pickl.ai/blog/advantages-and-disadvantages-random-forest/>
- Song, Y., Jiang, M., Li, S., & Zhao, S. (2024). Class-imbalanced financial distress prediction with machine learning: Incorporating financial, management, textual, and social responsibility features into index system. *Journal of Forecasting*, 43(3), 593–614. <https://doi.org/10.1002/for.3050>
- Sreedharan, M., Khedr, A. M., & El Bannany, M. (2020). A Comparative Analysis of Machine Learning Classifiers and Ensemble Techniques in Financial Distress Prediction.

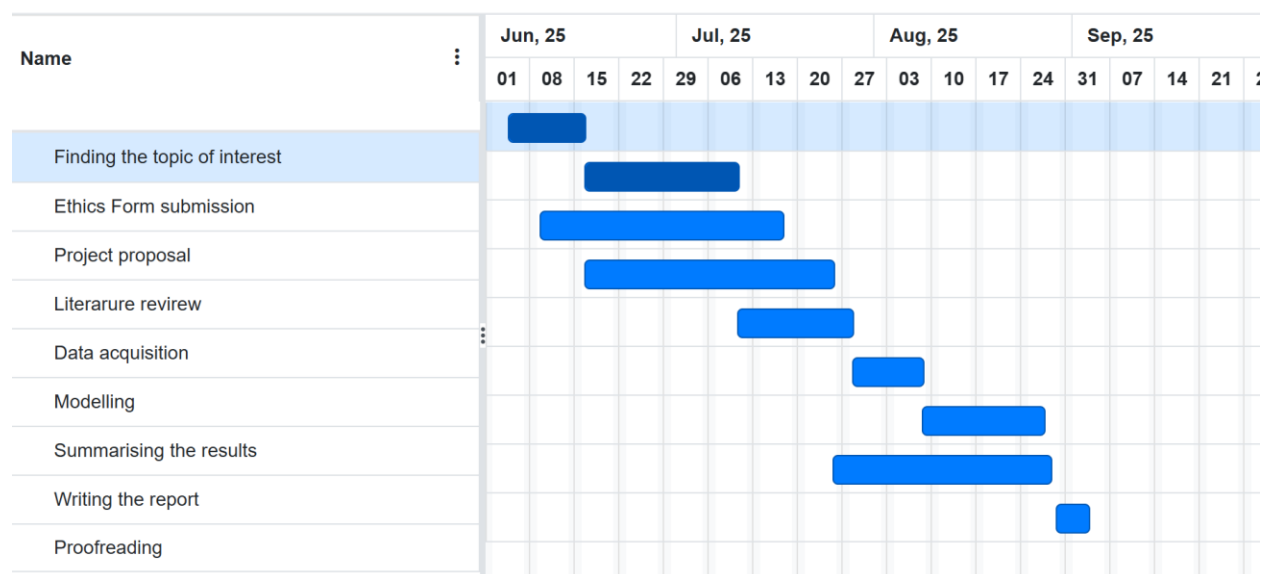
2020 17th International Multi-Conference on Systems, Signals & Devices (SSD), 653–657. <https://doi.org/10.1109/SSD49366.2020.9364178>

Colff, F., & Brümmer, L. (2019). Financial distress prediction using a machine learning model: A study of JSE-listed companies. *Management Dynamics*, 28(4), 29–40.

Ward, M. (2025, April 25). Retail sector in the UK. Houses of Commons Library. <https://researchbriefings.files.parliament.uk/documents/SN06186/SN06186.pdf>

Zizi, Y., Jamali-Alaoui, A., El Goumi, B., Oudgou, M., & El Moudden, A. (2021). An optimal model of financial distress prediction: A comparative study between neural networks and logistic regression. *Risks (Basel)*, 9(11), Article 200. <https://doi.org/10.3390/risks9110200>

Sugavasilakshmisahithi. (2025, February 23). Winsorization: a simple and effective way to handle outliers in your data. Medium. <https://medium.com/@sugavasilakshmisahithi/winsorization-a-simple-and-effective-way-to-handle-outliers-in-your-data-e9635c3bc049>



Time line of the project

